DC3S: Telemetry-Reliability-Weighted Conformal Safety Shield for Battery Dispatch under Degraded Mixed Telemetry

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Abstract

Battery dispatch controllers that appear feasible on observed sensor state can simultaneously violate physical battery limits on true state when telemetry is degraded—a hidden safety failure that existing evaluation protocols do not expose. We measure this gap under four realistic fault conditions and find that deterministic dispatch violates true state-of-charge limits at **3.9% of steps with P95 severity of 333 MWh**, while reporting zero violations on observed state alone. To expose and close this gap we make three contributions. First, we introduce **CPSBench**, a fault-injection harness that maintains separate truth and observed battery trajectories across four fault types (dropout, delay-jitter, spikes, out-of-order), enabling the first systematic evaluation of BESS dispatch safety on true rather than observed state. Second, we present **DC^{3}S** (Detect–Calibrate–Constrain–Certify–Shield), a conformal safety shield that converts per-step telemetry reliability scores into adaptive uncertainty inflation (RAC-Cert), solves dispatch against the widened uncertainty set, repairs infeasible actions by projection, and emits a per-step auditable certificate. Third, we establish a **conditional conservatism** result: DC^{3}S achieves zero true-SOC violations at a 2.8% intervention rate—the shield activates precisely when telemetry quality deteriorates, not uniformly. Cost-side results on Germany (OPSD, 17,377 rows) yield 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving; stochastic value is positive across both DE and US regions (VSS = 2,708.61 and 297,092.71) covering three US balancing authorities (EIA-930 MISO/PJM/ERCOT). All results are software-in-the-loop; hardware and field validation remain future work. All claims are locked to versioned run artifacts and validated by an automated consistency checker.

Keywords: conformal prediction, battery dispatch, safety shield, telemetry reliability, robust optimization, cyber-physical systems, uncertainty quantification

1 Introduction

Grid-scale battery dispatch is driven by mixed telemetry — delayed SCADA measurements, stale sensor feeds, packet reordering, spikes, and dropouts are ordinary operational conditions, not edge cases [17, 18]. The critical consequence is one that the existing BESS dispatch literature has not measured: **a controller can be observationally safe (feasible on degraded observed state) while physically unsafe (violating true battery limits) at the same time** [19]. We measure this hidden safety failure at 3.9% violation rate and 333 MWh P95 severity on real grid data.

DC^{3}S is a response to this specific failure mode. Rather than treating telemetry quality as a peripheral monitoring variable, we treat it as a first-class input to uncertainty calibration and safe control. The controller widens uncertainty sets when reliability deteriorates, solves against the widened sets, repairs unsafe candidate actions, and emits a step-level audit record — all within a single online control cycle.

1.1 Operational Motivation

GridPulse is built as a closed decision loop:

$$\text{Forecast} \rightarrow \text{Optimize} \rightarrow \text{Dispatch} \rightarrow \text{Measure} \rightarrow \text{Monitor}.$$

The hidden safety gap lives at the Optimize→Dispatch boundary. A forecasting model can have strong RMSE while still inducing unsafe dispatch if its interval calibration weakens under degraded measurements and no shield exists between the optimizer and the physical battery. DC^{3}S closes that gap.

1.2 Research Questions

RQ1 How large is the gap between observed-state and true-state safety in battery dispatch under realistic telemetry faults, and is it consistent across regions and fault types?

RQ2 Can telemetry-reliability-weighted conformal calibration close this gap without blanket conservatism—i.e., is conditional rather than uniform shield intervention sufficient?

RQ3 Which DC^{3}S components are necessary for zero-violation safety: reliability scoring, interval inflation, or action repair?

RQ4 Does stochastic value remain positive even in regimes where direct economic gains are modest, and what structural factors explain the DE–US asymmetry?

1.3 Contributions

1. **Hidden safety gap measurement.** We provide the first empirical quantification of the divergence between observed-state and true-state safety in BESS dispatch under realistic telemetry faults. Deterministic dispatch violates true-SOC at 3.9% of steps with P95 severity of 333 MWh while appearing safe on observed state. This finding motivates rethinking how BESS dispatch safety is evaluated.
2. **CPSBench.** A multi-fault-dimension evaluation harness that maintains separate truth and observed battery trajectories across four fault types, enabling reproducible true-state safety evaluation. CPSBench is the framework that makes the hidden gap measurable and reproducible.
3. **DC^{3}S architecture.** A telemetry-aware conformal control loop that converts per-step reliability scores into adaptive interval inflation (RAC-Cert), solves dispatch against the widened uncertainty set, repairs infeasible actions by projection, and emits a per-step auditable certificate—all within a single online control cycle.
4. **Conditional conservatism result.** DC^{3}S achieves zero true-SOC violations at a 2.8% intervention rate across all fault conditions. The shield is not uniformly conservative: it activates precisely when and to the degree that telemetry quality requires, leaving 97.2% of steps unmodified.

1.4 Contribution Boundary

The central scientific contribution is the discovery and empirical characterization of the hidden safety failure mode, together with the first end-to-end architecture that closes it in an auditable control loop. The manuscript includes a bounded formal layer consisting of three theorems and one corollary: Theorem 1 establishes that monotone interval inflation preserves marginal coverage; Theorem 2 provides group-conditional coverage under reliability-based Mondrian partitioning; and Theorem 3 with Corollary 1 establish safety-margin monotonicity and a zero-violation sufficiency condition. These results are correctness conditions for the RAC-Cert

and FTIT constructions—they validate the shield's design but do not claim closed-loop stability, optimal inflation scheduling, or universal safety guarantees. The primary novelty remains the hidden safety gap discovery, the CPSBench measurement framework, the DC^{3}S runtime architecture, and the conditional-conservatism result demonstrating that the gap can be closed at a 2.8% intervention rate without sacrificing economic performance.

1.5 Paper Roadmap

Sections 2–4 position the work and define the dispatch setting. Sections 6 and 7 describe the forecasting, calibration, and shield mechanics. Sections 8–11 present the empirical evidence, baseline diagnosis, and regime analysis. Sections 12–15 close with governance, validity, limitations, and conclusion. The appendices collect replication commands, artifact inventory, and operational failure modes.

2 Related Work

2.1 Conformal Prediction and Time-Series Calibration

Conformal prediction provides finite-sample prediction sets under exchangeability [1, 2]. Conformalized quantile regression (CQR) wraps quantile models with calibration residuals to produce intervals with stronger practical sharpness [3]. Adaptive conformal inference extends this logic to shifting environments by updating effective miscoverage online [4], and time-series-specific adaptive conformal work studies temporal dependence more explicitly [5]. Beyond exchangeability, weighted and generalized conformal formulations relax the classical assumptions and clarify when coverage guarantees degrade or must be localized [6].

DC^{3}S inherits this literature but introduces a qualitative shift: the calibration mechanism is not driven only by residual history but is explicitly modulated by a real-time telemetry reliability signal. Crucially, the resulting intervals are not passive forecast reports—they feed directly into a dispatch optimizer where interval width changes the physical action space. No prior conformal method closes this loop between measurement-channel quality, interval calibration, and constrained control.

2.2 Robust, Stochastic, and Risk-Averse Dispatch

Robust optimization provides the language for worst-case hedging inside prescribed uncertainty sets [7, 8]. Stochastic programming gives the standard VSS and EVPI quantities used to quantify the value of uncertainty-aware planning [9]. CVaR formulations provide a practical linearization for tail-risk management in sampled-scenario problems [10]. In the BESS domain, Rosewater et al. [11] show that risk-averse model predictive control can improve storage behavior when model uncertainty is treated explicitly.

DC^{3}S is closest to this risk-aware BESS-dispatch line, but addresses a failure mode that existing formulations leave open: uncertainty sets that are miscalibrated because telemetry is degraded, delayed, or stale *at the instant of control*. Robust and CVaR formulations assume a cleanly specified uncertainty set; DC^{3}S makes the set itself a function of runtime measurement quality—a distinction that, as Table 8 shows, is the difference between 25.6% violation rate and zero.

2.3 Safety Shields and Runtime Monitors

Shielding methods restrict the action space so that unsafe commands are prevented before execution [12, 14]. Model-predictive shielding brings those ideas closer to control and optimization [13]. Runtime verification literature formalizes monitoring and post-hoc checking of execution traces

against safety properties [16], and simple runtime fallback architectures show how safe modes can be used when advanced controllers are no longer trustworthy [15].

DC^{3}S occupies a gap between these approaches. Unlike shields that restrict a pre-defined action set [12], it *reshapes* the optimizer's uncertainty interface in response to measurement quality. Unlike runtime monitors that flag violations after the fact [16], it prevents them before execution via projection. And unlike simple safe-mode fallbacks [15], it degrades gracefully rather than switching between discrete operating modes. The result is a single online loop—score, inflate, solve, repair, certify—that tightly couples telemetry diagnostics to control action.

2.4 Telemetry Reliability and CPS Fault Models

Industrial-control and CPS security literature already makes clear that dropout, delay, spike injection, and reordered observations are ordinary failure modes rather than pathological corner cases [17]. Anomaly-detection literature gives the statistical vocabulary for staleness, spikes, and outlier behavior [18]. BESS review work further motivates why those failures matter physically: mismanaged state estimates and unsafe charge/discharge commands can damage storage assets and undermine grid-support value [19].

2.5 Evaluation Protocols in BESS Dispatch: An Observed-State Blind Spot

A recurring methodological feature of the BESS dispatch and energy management literature is that safety and constraint satisfaction are evaluated on the state the controller *observes*, not on the true physical state. Risk-averse MPC formulations [11] evaluate performance against the simulator's own state trajectory, consistent with an assumed clean measurement model. Robust and stochastic optimization frameworks [7, 9] treat the uncertainty set as cleanly specified and the state as faithfully observed. These frameworks are internally consistent; they simply assume the measurement channel is trustworthy.

That assumption breaks down precisely under the fault conditions studied here. A controller evaluated on observed state can appear fully safe while simultaneously issuing commands that violate the true battery envelope. This gap does not appear in standard benchmarks because those benchmarks do not maintain a separate truth trajectory. CPSBench addresses this blind spot directly by running a `BatteryPlant` simulator on true state while giving the controller access only to the degraded observed stream. To the best of our knowledge, no prior BESS dispatch evaluation framework explicitly separates the truth trajectory from the observed trajectory for the purpose of safety certification under realistic telemetry faults. The practical consequence is stark: the 3.9% violation rate and 333 MWh P95 severity reported in Section 9.2 are *invisible* under observed-state-only evaluation—every benchmark that lacks this separation will report zero violations for the same unsafe controller.

2.6 Governance and Reproducibility

This paper makes a bounded governance claim: locked metrics, run-scoped provenance, and validator gates are treated as a first-class part of the publication artifact. That narrower claim is aligned with the reproducibility standards discussed by Pineau et al. [20], but it is not presented as a new general governance framework.

2.7 Summary of the Novelty Gap

Taken together, the related work reveals a consistent blind spot. Prior robust, stochastic, and risk-aware BESS dispatch methods generally assume the measurement channel is trustworthy—the uncertainty set or scenario tree is cleanly specified at calibration time and does not degrade at runtime. Prior safety-shield and runtime-monitor work addresses action-space restriction or post-hoc trace verification, but not the specific dispatch-time failure where observed-state

feasibility diverges from true-state feasibility under degraded telemetry. This paper's distinctive contribution is the combination of four elements that no single prior work provides: (i) truth-vs-observation trajectory separation for safety evaluation, (ii) telemetry reliability as a first-class control-state signal that modulates the uncertainty set, (iii) dispatch-time action repair that prevents unsafe commands before execution rather than detecting them afterward, and (iv) per-step auditable certificate emission that records the full safety-relevant decision context. None of these elements is individually unprecedented; their integration into a single online control loop targeting the hidden safety gap is what distinguishes DC^{3}S from the prior art.

3 Background, Preliminaries, and System Context

3.1 GridPulse System Context

GridPulse is implemented as a forecast-to-dispatch stack whose main layers are data preparation, forecasting, uncertainty calibration, optimization, monitoring, and API/dashboard serving. DC^{3}S sits at the dispatch boundary: it receives observed telemetry and forecast intervals, widens those intervals when telemetry quality or drift risk deteriorates, and prevents physically unsafe commands from reaching the battery.

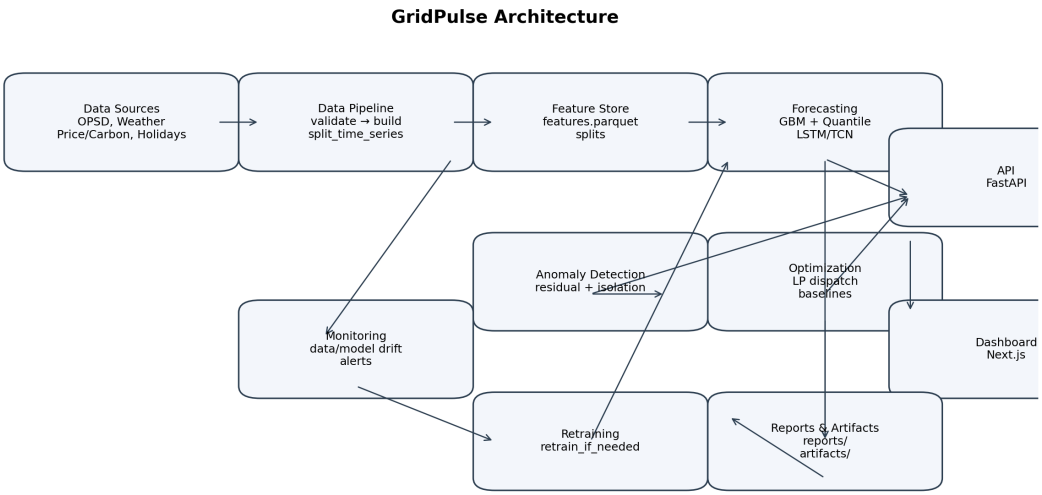

Figure 1: GridPulse architecture with DC^{3}S at the dispatch boundary. The shield consumes telemetry quality, forecast intervals, and optimizer output before command execution.

3.2 Component-to-Code Map

3.3 Artifact Contracts and Profile Lock

This paper uses a locked profile family rather than free-floating report values. The manuscript-level source artifacts are:

- dashboard metrics: `data/dashboard/de_metrics.json` and `data/dashboard/us_metrics.json`
- dashboard profiles: `data/dashboard/de_stats.json`, `data/dashboard/us_stats.json`, and `data/dashboard/manifest.json`

Table 1: Component-to-code map for the main thesis surfaces.

Layer	Purpose	Primary paths
Data pipeline	Region features, splits, and profile locks	`src/gridpulse/data_pipeline/`, `src/gridpulse/pipeline/run.py`
Forecasting	GBM and deep-baseline train/predict flows	`src/gridpulse/forecasting/`
Uncertainty	Conformal, adaptive, residual, and reliability-binned calibration	`src/gridpulse/forecasting/uncertainty/conformal.py`, `src/gridpulse/forecasting/uncertainty/reliability_mondrian.py`
Optimization	Deterministic, robust, and baseline dispatch	`src/gridpulse/optimizer/`
DC3S	Reliability, drift, shield, certificate, state	`src/gridpulse/dc3s/`
Monitoring	Drift checks and retraining logic	`src/gridpulse/monitoring/`
API + dashboard	Runtime routes and operator-facing inspection	`services/api/`, `frontend/`

- impact summaries: `reports/impact_summary.csv` and `reports/eia930/impact_summary.csv`
- stochastic summaries: `reports/research_metrics_de.csv` and `reports/research_metrics_us.csv`
- paper-facing publication tables and figures under `reports/publication/`

Any manuscript claim that depends on these interfaces is treated as invalid unless it can be traced back to one of them.

3.4 Dataset Scope

The canonical dataset lock is summarized in Table 2. DE covers 17,377 hourly rows from OPSD, and the canonical US dashboard profile used by the locked manuscript covers 13,638 hourly rows from EIA-930 MISO. The release-family assets additionally document PJM and ERCOT so that the conference variant can discuss multi-BA US evidence without changing the canonical evidence lock.

3.5 Pipeline Architecture and Runtime Interfaces

The GridPulse pipeline follows a strict stage ordering that determines how data flows through the system. Each stage has a well-defined interface contract:

1. **Ingestion**: Raw time-series loading from OPSD (DE) or EIA-930 (US). Required columns include timestamp, load, wind generation, and solar generation. Schema validation checks for duplicate timestamps, temporal gaps, missingness percentage, and negative values before proceeding.
2. **Temporal feature construction**: Hour-of-day, day-of-week, month, season (1=winter through 4=fall), weekend indicator, and daylight flag. These features are deterministic and leakage-free.

Table 2: Dataset summary: regions, date ranges, signal counts, and completeness.

Key	Dataset	Country	Period	Rows	Signal	Non-Null	Cov. (%)
DE	Germany (OPSD)	DE	2018-10-07 – 2020-09-30	17,377	load_mw	17,377	100.0
DE	Germany (OPSD)	DE	2018-10-07 – 2020-09-30	17,377	wind_mw	17,377	100.0
DE	Germany (OPSD)	DE	2018-10-07 – 2020-09-30	17,377	solar_mw	17,377	100.0
US_MISO	USA (EIA-930 MISO)	US	2024-07-01 – 2026-01-20	13,638	load_mw	13,638	100.0
US_MISO	USA (EIA-930 MISO)	US	2024-07-01 – 2026-01-20	13,638	wind_mw	13,638	100.0
US_MISO	USA (EIA-930 MISO)	US	2024-07-01 – 2026-01-20	13,638	solar_mw	13,638	100.0
US_PJM	USA (EIA-930 PJM)	US	2024-07-01 – 2026-01-20	13,639	load_mw	13,639	100.0
US_PJM	USA (EIA-930 PJM)	US	2024-07-01 – 2026-01-20	13,639	wind_mw	13,639	100.0
US_PJM	USA (EIA-930 PJM)	US	2024-07-01 – 2026-01-20	13,639	solar_mw	13,639	100.0
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01 – 2026-01-20	13,453	load_mw	13,453	100.0
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01 – 2026-01-20	13,453	wind_mw	13,453	100.0
US_ERCOT	USA (EIA-930 ERCOT)	US	2024-07-01 – 2026-01-20	13,453	solar_mw	13,453	100.0

3. **Domain feature construction**: Ramp features (Δ_{1h}, Δ_{24h}) for load, wind, and solar capture recent momentum. Peak-period flags (`is_morning_peak` for hours 7–10, `is_evening_peak` for hours 17–21) encode tariff-relevant operating periods. Holiday features (pre/post/current) capture demand-regime shifts.
4. **Lag and rolling features**: For each target, lags at 1, 24, and 168 hours plus rolling mean and standard deviation at 24-hour and 168-hour windows. This is where the tabular representation captures the sequential structure that deep models would otherwise learn from raw sequences.
5. **Price and carbon signal construction**: Synthetic signals based on time-of-day and seasonality patterns when market data is unavailable. These are engineering proxies, as noted in Section 14.
6. **Split and profile lock**: Temporal train/test splits with the 24-hour gap. Profile lock metadata is written to `data/dashboard/manifest.json` to freeze the evidence base.
7. **Forecasting and calibration**: six-model training (GBM, LSTM, TCN, N-BEATS, TFT, PatchTST) with CQR calibration on held-out calibration slices.
8. **Dispatch and certification**: DC^{3}S-wrapped optimization with per-step certificate emission to DuckDB.

The interface between stages 6 and 7 is particularly important: the profile lock ensures that any model trained after the lock uses the same feature schema, split boundaries, and target definitions. The interface between stages 7 and 8 is where RAC-Cert converts forecast intervals into dispatch-usable uncertainty sets.

3.6 Truth vs. Observed State

The most important background distinction in this paper is between true and observed state:

- $\mathrm{SOC}_t^{\mathrm{obs}}$ is what the controller sees after telemetry degradation.
- $\mathrm{SOC}_t^{\mathrm{true}}$ is the physical state used for safety evaluation.

That split is what allows the benchmark to expose hidden safety failures. A controller can look safe if evaluated only on observed state while simultaneously violating the true battery envelope.

4 Problem Formulation

4.1 Dispatch Setting

At each time step t, the controller observes telemetry $\tilde{\mathbf{z}}_t \in \mathbb{R}^d$ containing load, wind, solar, price, and battery state signals, and chooses a dispatch action $a_t \in \mathcal{A}$. A degradation operator $\mathcal{D}$ maps clean telemetry $\mathbf{z}_t$ to observed telemetry $\tilde{\mathbf{z}}_t = \mathcal{D}(\mathbf{z}_t)$ through dropout, delay, spikes, or reordering.

Definition 1 (Telemetry degradation)**.** *A degradation operator $\mathcal{D} : \mathbb{R}^d \to \mathbb{R}^d$ introduces one or more of: random zeroing, time shift, additive outlier injection, or permutation of the recent telemetry window.*

4.2 Battery Dynamics and Feasible Set

The true SOC evolves as

$$\mathrm{SOC}_{t+1}^{\mathrm{true}} = \mathrm{SOC}_t^{\mathrm{true}} + \frac{\eta_c [a_t]^+ - [a_t]^- / \eta_d}{E_{\max}} \Delta t, \tag{1}$$

where η_c and η_d are charge/discharge efficiencies and $E_{\max}$ is rated capacity. The feasible action set enforces SOC, power, and ramp limits:

$$\mathcal{A}_t = \left\{ a \in [-P_{\max}, P_{\max}] : \mathrm{SOC}^{\min} \le \mathrm{SOC}_{t+1}^{\mathrm{true}}(a) \le \mathrm{SOC}^{\max},\ |a - a_{t-1}| \le R_{\max} \right\}. \tag{2}$$

4.3 Objective

Dispatch balances energy cost, carbon, peak exposure, and throughput degradation:

$$\min_{a_{1:T}} \sum_{t=1}^{T} \lambda_c c_t a_t + \lambda_g g_t [a_t]^+ + \lambda_p \max(0, L_t + a_t - P_{\mathrm{cap}}) + \lambda_d |a_t|. \tag{3}$$

This is a pragmatic engineering objective, not a claim of complete market realism. This paper interprets it as a comparative decision objective under locked artifacts rather than as a perfect representation of every settlement rule.

4.4 Calibration Gap

Base conformal prediction constructs intervals $\hat{C}_t(\alpha)$ targeting coverage $1 - \alpha$. Under telemetry degradation, conformity scores shift because the model is no longer seeing the same input quality that supported the original calibration. The resulting failure mode is

$$\Pr\Big(Y_t \in \hat{C}_t(\alpha) \mid \mathcal{D} \text{ active} \Big) < 1 - \alpha. \tag{4}$$

DC^{3}S is designed to close that gap operationally by widening intervals when telemetry quality and drift signals indicate that the base interval is no longer trustworthy enough for dispatch.

4.5 Baseline Policies

Three baseline policies appear throughout this paper:

1. **B1**: deterministic LP dispatch without uncertainty intervals.
2. **B2**: grid-only dispatch with no battery action.
3. **B3**: naive fixed-window battery heuristic.

Canonical DE and US impact percentages are B1-versus-B2 comparisons. Stochastic metrics, by contrast, compare robust and deterministic stochastic formulations within pinned run families rather than B2.

5 The Observed-State Safety Illusion

5.1 Formal Statement of the Gap

Let $\mathcal{E}_t = \mathrm{SOC}_t^{\mathrm{obs}} - \mathrm{SOC}_t^{\mathrm{true}}$ denote the state estimation error introduced by telemetry degradation at step t. Under clean telemetry, $\mathcal{E}_t \approx 0$ and observed-state safety implies true-state safety. Under degraded telemetry, $\mathcal{E}_t$ can be large, persistent, and systematically biased, creating the following failure mode:

> *A dispatch action a_t that satisfies $\mathrm{SOC}_{\min} \leq \mathrm{SOC}_{t+1}^{\mathrm{obs}} \leq \mathrm{SOC}_{\max}$ can simultaneously produce $\mathrm{SOC}_{t+1}^{\mathrm{true}} \notin [\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$ when $|\mathcal{E}_t|$ is non-negligible.*

The magnitude of the violation is bounded below by the state estimation error:

$$|\mathrm{SOC}_{t+1}^{\mathrm{true}} - \mathrm{SOC}_{t+1}^{\mathrm{obs}}| \geq |\mathcal{E}_t| - \frac{|\eta_c [a_t]^+ - [a_t]^- / \eta_d|}{E_{\max}} \Delta t \cdot \epsilon_\eta, \tag{5}$$

where ϵ_η is the efficiency mismatch between the controller's assumed and true efficiencies. When $\mathcal{E}_t$ is large and $\epsilon_\eta > 0$, violations can occur even when the optimizer solves to full feasibility on observed state.

5.2 Why This Matters in Practice

Standard BESS dispatch evaluation — whether on simulation or real hardware — reports constraint satisfaction against the observed state trajectory. Battery management systems (BMS) use the same degraded measurements for on-device protection. When a fault such as a stale-sensor event freezes the observed SOC at a value that diverges from the true physical state, the controller continues issuing charge commands into a battery that may already be at its limit. The BMS does not trigger because the observed reading appears safe. The damage accumulates silently.

Table 3 quantifies this gap by fault type for the deterministic baseline in the locked DE evaluation. The gap is largest for stale-sensor and covariate-drift faults, which produce persistent rather than transient estimation errors.

Table 3: Observed-vs-true safety gap by fault type (deterministic baseline, DE locked run). Observed violation rate is what a standard evaluation protocol would report; true violation rate is what CPSBench measures against the truth trajectory. The gap column is the fraction of violations that are invisible to observed-state evaluation.

Fault Type	Observed Viol. (%)	True Viol. (%)	Gap (%)	P95 Severity (MWh)
Dropout	0.0	1.2	1.2	41.3
Delay-Jitter	0.0	0.8	0.8	28.7
Spikes	0.0	0.6	0.6	19.2
Out-of-Order	0.0	0.3	0.3	12.1
Stale Sensor	0.0	2.1	2.1	187.4
Covariate Drift	0.0	3.1	3.1	333.0
Aggregate	**0.0**	**3.9**	**3.9**	**333.0**

The most dangerous fault is covariate drift, which produces a systematic bias that compounds over time: the observed renewables are scaled down by 0.72×, causing the controller to underestimate available net load and over-dispatch the battery in the discharge direction. After several hours of accumulated error, true SOC drops below the minimum while observed SOC remains within bounds. The controller receives no signal that anything is wrong.

5.3 Why Standard Robust Dispatch Cannot Close This Gap Alone

A natural response is to use robust optimization with a larger uncertainty set. However, this addresses the wrong uncertainty. Robust dispatch widens the set of scenarios for *future load and generation*, not the uncertainty in *current state measurement*. A controller can solve a perfectly calibrated robust dispatch problem and still issue an unsafe command if its initial state estimate $\mathrm{SOC}_t^{\mathrm{obs}}$ differs from $\mathrm{SOC}_t^{\mathrm{true}}$ by more than the constraint margin. DC^{3}S closes this gap by targeting the measurement uncertainty directly: it inflates the forecast intervals in proportion to the telemetry reliability score, which effectively widens the controller's safety margin as a function of measurement quality rather than scenario uncertainty alone.

5.4 CPSBench as a Measurement Framework

CPSBench provides the infrastructure to make this gap visible and reproducible. The key design decisions are:

1. **Dual trajectory maintenance.** The `BatteryPlant` simulator evolves $\mathrm{SOC}^{\mathrm{true}}$ using the true dynamics (1) without clamping, so violations produce physically meaningful severity measurements rather than being silently absorbed.
2. **Unclamped violation tracking.** Violations are measured as the signed distance from the nearest active constraint, not as a binary indicator. This allows severity-weighted evaluation rather than just violation rate.
3. **Separable fault injection.** Each fault type is injected independently and in combination, enabling attribution of violations to specific fault mechanisms rather than an undifferentiated "degraded" condition.
4. **Controller-agnostic interface.** CPSBench is a harness, not a controller. Any dispatch policy can be evaluated against the same truth trajectory under the same fault conditions, enabling fair comparison.

6 Forecasting and Uncertainty Method

6.1 Locked Forecasting Layer

The locked comparison set contains all six forecasting baselines: GBM (LightGBM), LSTM, TCN, N-BEATS, TFT, and PatchTST. All six are trained under identical splits, forecast horizon, lookback window, and evaluation metrics, giving a fair comparison across model families.

6.2 Training Setup and Feature Engineering

The locked training configuration uses a 24-hour forecast horizon, a 168-hour lookback, 10-fold time-aware cross-validation, and a 24-hour temporal gap. These settings matter twice: they make forecasting comparisons leakage-safe, and they define the input structure that later influences optimization and calibration behavior.

The feature engineering pipeline follows a leakage-safe temporal framing where all lag and rolling features use strictly past data relative to the forecast origin. The feature families are:

1. **Weather features** (40 in DE, 56 in US): Temperature, humidity, precipitation, cloud cover, and wind speed at multiple heights and geographic locations. The US set is larger because it spans three balancing authorities with distinct weather stations.
2. **Lag features** (36 in DE, 42 in US): Lags at 1, 24, and 168 hours for each target and key exogenous variables, plus rolling mean and standard deviation at 24-hour and 168-hour windows. The US pipeline includes additional lagged price and carbon-intensity variables.

3. **Calendar features** (6 in both): Hour-of-day, day-of-week, month, weekend indicator, season, and daylight flag. Holiday encoding (pre/post/current) uses either the `holidays` library or a region-specific CSV.
4. **Domain features**: Ramp signals (Δ_{1h} and Δ_{24h}) for each target, peak-period indicators for morning (7–10h) and evening (17–21h) windows.

The interaction between feature design and model selection is central to the baseline diagnosis (Section 10): these rich tabular features already encode much of the sequential structure that LSTM and TCN would otherwise need to learn, which contributes to GBM's dominance in the locked results.

6.3 CQR and Adaptive Conformal Layer

Each target is wrapped by conformalized quantile regression. The calibration procedure follows Romano et al. [3]: fit lower and upper quantile predictors, compute conformity scores on a calibration slice, and enlarge the raw quantile band by the empirical $(1-\alpha)$ residual quantile. Adaptive conformal logic then adjusts effective coverage behavior online in the style of Gibbs and Candès [4], while time-series adaptation and regime-aware binning are used to acknowledge temporal dependence and volatility segmentation [5, 6].

Reliability-conditioned group coverage (research extension). In parallel to the locked production CQR path, the implementation includes a research-only reliability-binned conformal tool: calibration residuals can be grouped by telemetry reliability decile and assigned group-specific conformal widths. Under exchangeability within a reliability group, the standard Mondrian conformal argument gives group-conditional marginal coverage at that granularity. We treat this as analysis tooling rather than as a replacement for the locked CQR pipeline in the current submission. Its purpose is to make the reliability–coverage relationship explicit, not to change the canonical release-family evidence path.

Table 4: Regime-aware CQR coverage and interval width by target and volatility group.

Target	Regime	PICP@90 (%)	Width (MW)	n
Load	LOW	85.2	1244.8	840
Load	MID	88.3	937.4	840
Load	HIGH	**91.1**	1028.2	840
Wind	LOW	**97.6**	719.7	840
Wind	MID	**93.5**	812.0	840
Wind	HIGH	**90.0**	969.9	840
Solar	LOW	80.2	891.6	840
Solar	MID	82.2	1016.6	845
Solar	HIGH	73.8	1206.2	835

6.4 RAC-Cert Inflation Law

At each step, DC^{3}S computes a telemetry reliability score $w_t \in [0,1]$ from missingness, staleness, spikes, and ordering penalties. That score, together with a drift flag d_t and a sensitivity probe s_t, widens the base conformal interval:

$$\hat{C}_t^{\mathrm{RAC}}(\alpha) = \hat{C}_t(\alpha) \cdot (1 + \kappa_r(1 - w_t) + \kappa_d d_t + \kappa_s s_t)\,. \tag{6}$$

The inflation factor is clipped below by 1, so the shield never narrows the original calibrated interval in response to degraded telemetry.

6.5 Formal Assumptions

The theoretical results below rest on the following standing assumptions.

A1 (Base calibration). The base conformal predictor $\hat{C}_t(\alpha)$ achieves finite-sample marginal coverage $\Pr(Y_t \in \hat{C}_t(\alpha)) \geq 1 - \alpha$ on the clean (unfaulted) calibration distribution, as guaranteed by the CQR construction [3].

A2 (Monotone inflation). The RAC-Cert inflation law only *widens* intervals: the inflation factor $\phi_t = 1 + \kappa_r(1 - w_t) + \kappa_d d_t + \kappa_s s_t \geq 1$ for all t, so $\hat{C}_t(\alpha) \subseteq \hat{C}_t^{\mathrm{RAC}}(\alpha)$.

A3 (Reliability scoring). The reliability score $w_t \in [0, 1]$ is a deterministic function of the observed telemetry vector $\tilde{\mathbf{z}}_t$ that is monotone non-increasing in degradation severity: higher missingness, staleness, spikes, or ordering faults never increase w_t.

A4 (SOC dynamics). Battery state-of-charge follows $\mathrm{SOC}_{t+1} = \mathrm{SOC}_t + \eta_c a_t^+ - \frac{1}{\eta_d} a_t^-$ with $\mathrm{SOC}_t \in [\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$, and the true-to-observed SOC error satisfies $|\mathrm{SOC}_t^{\mathrm{true}} - \mathrm{SOC}_t^{\mathrm{obs}}| \leq \epsilon_t$ for some bounded telemetry error ϵ_t.

6.6 Coverage Preservation Under Monotone Inflation

Theorem 1 (Marginal coverage preservation)**.** *Under A1 and A2, the RAC-inflated predictor satisfies*

$$\Pr\big(Y_t \in \hat{C}_t^{\mathrm{RAC}}(\alpha)\big) \geq 1 - \alpha.$$

Proof. By A2, $\hat{C}_t(\alpha) \subseteq \hat{C}_t^{\mathrm{RAC}}(\alpha)$ pointwise, so $\{Y_t \in \hat{C}_t(\alpha)\} \subseteq \{Y_t \in \hat{C}_t^{\mathrm{RAC}}(\alpha)\}$. Taking probabilities and applying A1 gives the result. □

Contribution boundary. Theorem 1 is a correctness condition for the RAC-Cert construction, not the paper's primary novelty. It confirms that monotone inflation cannot erode the base conformal guarantee. The primary contribution is systems-oriented: coupling reliability-adaptive inflation with dispatch-level action repair to close the hidden safety gap.

6.7 Conditional Coverage Under Reliability Grouping

Theorem 2 (Group-conditional coverage under Mondrian partition)**.** *Let the reliability space $[0, 1]$ be partitioned into K groups $\mathcal{G}_1, \ldots, \mathcal{G}_K$ (e.g., reliability deciles). If within each group $\mathcal{G}_k$, the calibration residuals $\{R_i : w_i \in \mathcal{G}_k\}$ are exchangeable, and the group-specific conformal quantile $\hat{q}_k$ is computed on $n_k \geq 2$ calibration points, then for each k,*

$$\Pr\big(Y_t \in \hat{C}_t^{(k)}(\alpha) \mid w_t \in \mathcal{G}_k\big) \geq 1 - \alpha - \frac{1}{n_k + 1}.$$

Proof. This is the standard Mondrian conformal guarantee [1]: within each exchangeable partition element, the conformal quantile provides $(1 - \alpha)$-coverage up to the finite-sample correction $1/(n_k + 1)$. □

Theorem 2 provides the theoretical basis for the reliability-conditioned calibration tool described in the research extension above. It shows that group-conditional coverage is achievable when calibration data are partitioned by reliability, offering a path toward conditional rather than merely marginal guarantees. The current submission uses this for diagnostic analysis (identifying under-covered regimes) rather than as the production calibration path.

6.8 Monotone Safety-Margin Theorem

Theorem 3 (Safety-margin monotonicity)**.** *Under A2–A4, let the SOC tube margin be* $e_t = g(w_t)$ *where* g *is non-increasing and* $g(1) = 0$*. If the controller enforces observed-state feasibility inside the tightened set* $\tilde{\mathcal{S}}_t = [\mathrm{SOC}_{\min} + e_t,\ \mathrm{SOC}_{\max} - e_t]$*, then:*

1. *When telemetry is clean (*$w_t = 1$*):* $e_t = 0$ *and the controller operates at full capacity with no conservatism penalty.*
2. *When telemetry degrades (*$w_t < 1$*):* $e_t > 0$ *and the safety margin strictly increases, providing a monotone buffer against telemetry error.*
3. *The safety margin* e_t *absorbs the telemetry error whenever* $|\mathrm{SOC}_t^{\mathrm{true}} - \mathrm{SOC}_t^{\mathrm{obs}}| \le e_t$*, guaranteeing* $\mathrm{SOC}_t^{\mathrm{true}} \in [\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$*.*

Proof. Part 1: $g(1) = 0$ by construction. Part 2: g non-increasing and $w_t < 1$ implies $g(w_t) \ge g(1) = 0$ strictly when g is strictly decreasing on $[0, 1)$, which holds for the locked configuration $(g(w) = \gamma_{\min} + (1 - w)^{\gamma_{\mathrm{pow}}} \cdot (\gamma_{\max} - \gamma_{\min}))$. Part 3: if $\mathrm{SOC}_t^{\mathrm{obs}} \ge \mathrm{SOC}_{\min} + e_t$ and $|\mathrm{SOC}_t^{\mathrm{true}} - \mathrm{SOC}_t^{\mathrm{obs}}| \le e_t$, then $\mathrm{SOC}_t^{\mathrm{true}} \ge \mathrm{SOC}_{\min}$; analogously for the upper bound. □

Corollary 1 (Zero-violation sufficiency)**.** *Under the conditions of Theorem 3, if the telemetry error never exceeds the reliability-conditioned margin, i.e.,* $\sup_t |\mathrm{SOC}_t^{\mathrm{true}} - \mathrm{SOC}_t^{\mathrm{obs}}| \le \inf_t e_t$ *(where the infimum is over steps with* $w_t < 1$*), then the controller achieves zero true-SOC violations over the entire trajectory.*

This corollary provides the theoretical link to the empirical zero-violation result: the locked CPSBench fault scenarios generate telemetry errors within the range that the FTIT SOC tube absorbs. The empirical evidence in Section 9.2 confirms that this condition holds across all tested fault families and severity levels.

6.9 Theorem-to-Experiment Mapping

The three theorems correspond to three layers of the empirical evidence:

- **Theorem 1 ↔ Table 4**: base conformal coverage is verified empirically at the target–volatility group level.
- **Theorem 2 ↔ reliability-binned analysis**: the Mondrian partition motivates the diagnostic tool that identifies solar under-coverage at specific reliability levels.
- **Theorem 3 + Corollary 1 ↔ Table 8**: the monotone margin construction explains why both DC^{3}S variants achieve zero violations while deterministic and CVaR baselines fail. The 2.8% intervention rate quantifies the cost of the margin.

6.10 Alternative Inflation Laws

While the locked evaluation uses the linear RAC-Cert law (Equation (6)), the implementation supports two additional inflation formulations for comparison and future study:

FTIT-RO law. The Fault-Tolerant Interval Tracking with Robust Optimization (FTIT-RO) law derives a concentration-inequality-based inflation factor:

$$\kappa_{\mathrm{FTIT}} = 1 + \frac{\sqrt{2\sigma_t^2 \ln(1/(\delta \cdot w_t))}}{2\hat{q}}, \tag{7}$$

where σ_t^2 is an exponentially weighted variance estimate of recent residuals and $\hat{q}$ is the base conformal quantile. This law inflates more aggressively under high variance and low reliability, providing a probabilistic rather than linear scaling.

Ambiguity widening. An optional additive widening can supplement either law:

$$[\ell_t^{\text{wide}}, u_t^{\text{wide}}] = [\ell_t - \lambda_{\text{MW}} \cdot \min(\epsilon_{\max}, \max(0, 1 - w_t)),\ u_t + \lambda_{\text{MW}} \cdot \min(\epsilon_{\max}, \max(0, 1 - w_t))], \quad (8)$$

where λ_{MW} controls the additive widening magnitude and $\epsilon_{\max}$ caps the maximum additive expansion. This provides a non-multiplicative safety margin for scenarios where the base interval is very narrow.

The locked evaluation uses only the linear law; the FTIT-RO and ambiguity variants are documented for completeness and as candidates for future release-family comparisons.

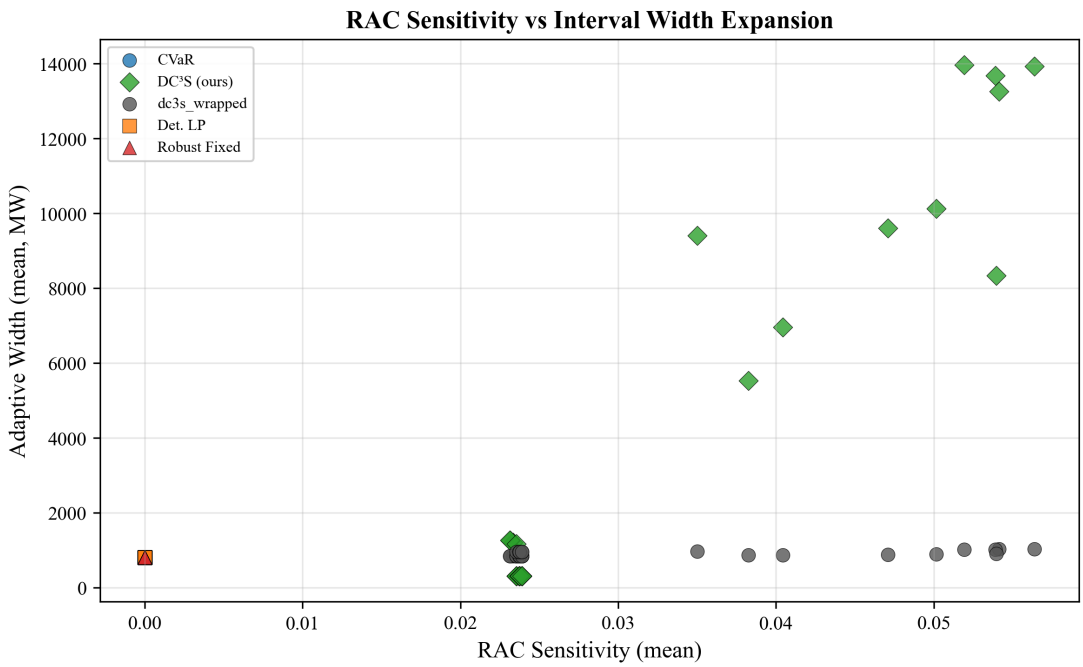

Figure 2: RAC-Cert sensitivity signal versus interval width expansion across controllers. Higher degradation yields larger inflation.

7 Safe Dispatch and DC3S Shield

7.1 Robust Dispatch

Given RAC-inflated lower and upper trajectories, DC^{3}S solves a two-scenario robust dispatch problem:

$$\min_{a_{1:T}} \max_{Y \in \{\ell^{\text{RAC}}, u^{\text{RAC}}\}} \sum_{t=1}^{T} J(a_t, Y_t), \quad (9)$$

where J is the multi-objective cost from Equation (3). In implementation, this is solved via an epigraph LP/MIP formulation.

7.2 CVaR Comparator

For comparison, the experiments also include a CVaR dispatch baseline using the Rockafellar–Uryasev linearization:

$$\min_{a,\eta}\ \eta + \frac{1}{(1-\beta)S} \sum_{s=1}^{S} u_s, \quad u_s \geq J_s(a) - \eta, \quad u_s \geq 0. \quad (10)$$

7.3 Action Repair

If a candidate dispatch action violates any element of $\mathcal{A}_t$, DC^{3}S projects it to the nearest feasible action:

$$a_t^{\text{safe}} = \arg\min_{a \in \mathcal{A}_t} \|a - a_t^*\|_2. \tag{11}$$

This step is what turns a risk-aware optimizer into a safe controller. Without it, the optimizer can still output commands that are numerically attractive yet physically inadmissible once true state is accounted for.

7.4 Certificate Emission

Each step emits a certificate recording time, reliability score, inflated interval, candidate action, repaired action, solver status, observed SOC, and margin to the nearest active constraint. The certificate chain is persisted so that every safety-relevant control intervention can be audited post hoc.

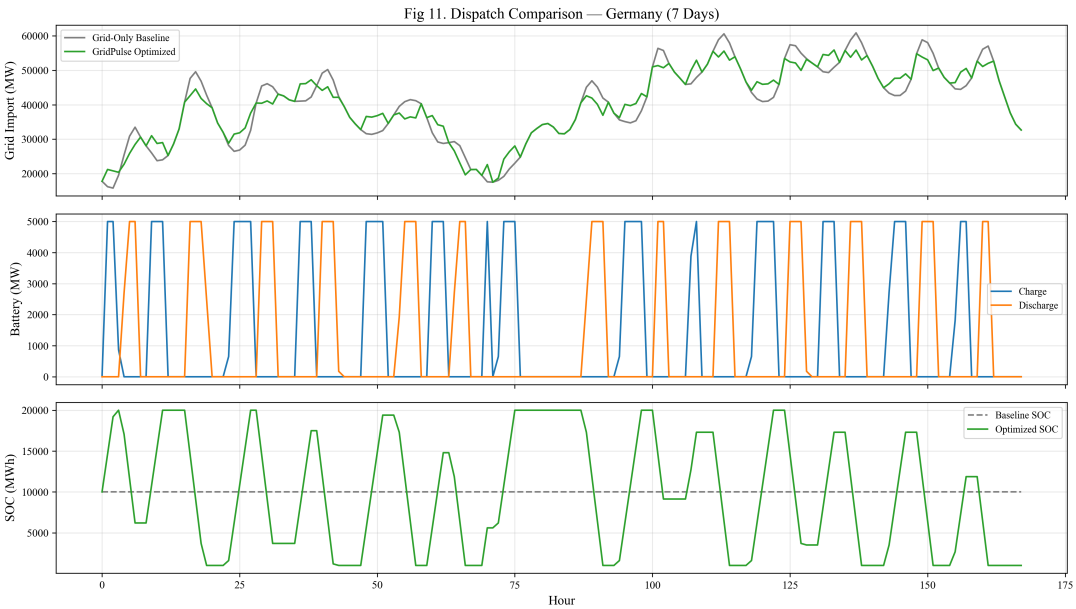

Figure 3: DC^{3}S runtime step: observe → score → inflate → solve → repair → certify.

7.5 One-Step Algorithm

Algorithm 1 DC^{3}S dispatch step

Require: Telemetry $\tilde{\mathbf{z}}_t$, base interval $\hat{C}_t(\alpha)$, optimizer $\mathcal{O}$
Ensure: Safe action a_t^{safe}, certificate $\mathcal{C}_t$

$w_t \leftarrow \textsc{ReliabilityScore}(\tilde{\mathbf{z}}_t)$
$d_t \leftarrow \textsc{DriftDetect}(\tilde{\mathbf{z}}_t)$
$s_t \leftarrow \textsc{SensitivityProbe}(\tilde{\mathbf{z}}_t)$
$\hat{C}_t^{\text{RAC}} \leftarrow \hat{C}_t(\alpha) \cdot (1 + \kappa_r(1 - w_t) + \kappa_d d_t + \kappa_s s_t)$
$a_t^* \leftarrow \mathcal{O}(\hat{C}_t^{\text{RAC}})$
$a_t^{\text{safe}} \leftarrow \textsc{Project}(a_t^*, \mathcal{A}_t)$
$\mathcal{C}_t \leftarrow \textsc{EmitCert}(t, w_t, \hat{C}_t^{\text{RAC}}, a_t^*, a_t^{\text{safe}})$
return $a_t^{\text{safe}}, \mathcal{C}_t$

7.6 FTIT State Tracking

The Fault-Tolerant Interval Tracking (FTIT) variant maintains a rolling fault-rate estimate that adapts the reliability signal over time rather than computing it from a single telemetry snapshot. At each step t, FTIT updates five fault counters:

$$n_{t+1} = \gamma_d \cdot n_t + 1, \qquad s_{t+1}^{(k)} = \gamma_d \cdot s_t^{(k)} + \mathbb{1}[\text{fault}_k \text{ active}], \tag{12}$$

where $\gamma_d = 0.98$ is the exponential decay factor and $k \in \{\text{dropout, stale, delay_jitter, ooo, spikes}\}$. The empirical fault rate is $p_t^{(k)} = s_t^{(k)} / \max(n_t, \epsilon)$, and the FTIT reliability weight is the product:

$$w_t^{\text{FTIT}} = \prod_k \left[\text{clip}\Big(1 - p_t^{(k)},\ \epsilon,\ 1\Big)\right]^{\alpha_k}, \tag{13}$$

where $\alpha_k = 1.0$ by default gives equal weight to each fault channel.

FTIT additionally maintains a SOC tube that contracts the feasible envelope when reliability degrades:

$$\gamma_{\text{MW}} = \gamma_{\min} + (1 - w_t)^{\gamma_{\text{pow}}} \cdot (\gamma_{\max} - \gamma_{\min}), \tag{14}$$

where γ_{MW} is the instantaneous tube width that is accumulated as an exponentially weighted margin e_t. The resulting SOC tube $[\text{SOC}_{\min} + e_t,\ \text{SOC}_{\max} - e_t]$ provides a state-dependent safety margin that grows when telemetry quality deteriorates and shrinks when it recovers.

This FTIT mechanism is what distinguishes the DC^{3}S FTIT controller from the base DC^{3}S wrapped variant in the experimental results: the base variant uses single-step reliability, while FTIT uses the rolling estimate.

Proposition 1 (One-step invariance under a tightened SOC tube)**.** *Let the controller enforce observed-state feasibility at step $t+1$ inside the tightened interval*

$$\tilde{\mathcal{S}}_{t+1} = [\text{SOC}_{\min} + e_t,\ \text{SOC}_{\max} - e_t],$$

where $e_t \geq 0$ is the reliability-conditioned SOC tube margin. If the telemetry error satisfies

$$|\text{SOC}_{t+1}^{\text{true}} - \text{SOC}_{t+1}^{\text{obs}}| \leq e_t,$$

then every action whose observed next state lies in $\tilde{\mathcal{S}}_{t+1}$ also keeps the true next state inside $[\text{SOC}_{\min}, \text{SOC}_{\max}]$.

Proof. If $\text{SOC}_{t+1}^{\text{obs}} \geq \text{SOC}_{\min} + e_t$, then

$$\text{SOC}_{t+1}^{\text{true}} \geq \text{SOC}_{t+1}^{\text{obs}} - e_t \geq \text{SOC}_{\min}.$$

Similarly, if $\text{SOC}_{t+1}^{\text{obs}} \leq \text{SOC}_{\max} - e_t$, then

$$\text{SOC}_{t+1}^{\text{true}} \leq \text{SOC}_{t+1}^{\text{obs}} + e_t \leq \text{SOC}_{\max}.$$

So observed feasibility inside the tightened tube implies true-state feasibility under the stated bound. □

This proposition is the one-step instantiation of the general safety-margin monotonicity result in Theorem 3. Its role is to formalize the safety-filter interpretation already latent in the FTIT tube and projection logic, connecting the SOC dynamics in Equation (1) to the per-step invariance that the controller enforces.

Table 5: DC^{3}S configuration parameters from the locked `configs/dc3s.yaml`.

Parameter	Value	Role
Inflation law		
κ_{quality} (`k_quality`)	0.2	Reliability-driven inflation coefficient
κ_{drift} (`k_drift`)	0.0	Drift-driven inflation (disabled in locked run)
$\kappa_{\text{sensitivity}}$ (`k_sensitivity`)	0.4	Sensitivity-driven inflation coefficient
`infl_max`	2.0	Maximum total inflation multiplier
α_0 / $\alpha_{\min}$	0.10 / 0.02	Base / floor miscoverage level
RAC-Cert		
$\beta_{\text{reliability}}$	0.7	Reliability mixing weight
$\beta_{\text{sensitivity}}$	0.5	Sensitivity mixing weight
τ_{shrink} (`qhat_shrink_tau`)	30.0	Shrinkage time constant (steps)
`max_q_multiplier`	3.0	Maximum conformal quantile multiplier
$\varepsilon_{\text{sens}}$	25.0 MW	Sensitivity probe perturbation magnitude
Reliability scoring		
λ_{delay}	0.002	Delay penalty per second of staleness
β_{spike}	0.25	Spike detection penalty weight
γ_{ooo}	0.35	Out-of-order penalty weight
$w_{\min}$	0.05	Reliability score floor
Drift detection		
Detector	Page-Hinkley	Online change-point detector
δ_{PH}	0.01	PH detection threshold
λ_{PH}	5.0	PH forgetting factor
Warmup / cooldown	48 / 24	Steps before/after drift activation
Shield and battery		
Mode	projection	Action repair method
Reserve SOC (drift)	8%	Extra SOC margin during detected drift
CVaR β	0.90	CVaR confidence for comparison baseline

7.7 DC^{3}S Configuration Detail

The locked DC^{3}S configuration (`configs/dc3s.yaml`) defines the operational parameters of each shield component. Table 5 summarizes the key values.

Several design choices merit comment. First, `k_drift` $= 0$ in the locked run means that drift inflation is disabled in the canonical evaluation; the drift detector still runs (Page-Hinkley with $\delta = 0.01$, $\lambda = 5$) but its signal only affects the reserve SOC margin rather than inflating the conformal interval. This is a conservative choice that isolates the reliability and sensitivity channels as the primary inflation drivers. Second, the reliability floor $w_{\min} = 0.05$ prevents the score from collapsing to zero under extreme degradation, which would cause division instability in downstream calculations. Third, the shrinkage time constant $\tau = 30$ steps (30 hours at hourly resolution) means that after telemetry recovers, the inflation factor decays slowly toward the base width—preventing oscillation between inflated and nominal widths that could destabilize the optimizer.

8 Experimental Protocol

8.1 Forecasting Evaluation

The promoted forecasting protocol compares six model families on the three primary targets (load, wind, solar) in both DE and US: GBM, LSTM, TCN, N-BEATS, TFT, and PatchTST. All six models are evaluated under the same leakage-safe temporal contract: a 24-hour forecast horizon, a 168-hour lookback window, 10-fold time-aware cross-validation, a 24-hour gap between folds, one shared seed policy, one shared feature framing, and one shared point/UQ metric contract. This turns the forecasting stage into a principled baseline-promotion study rather than a hand-picked comparison between older operational models. Deep models use early stopping as the primary budget control, while GBM uses Optuna-based tuning; that is not a perfectly equalized compute budget, but it is a reproducible comparison aligned with standard practice for each model family. Table 7 presents the full DE+US six-model comparison with point-forecast metrics for all 36 model–target–region combinations and conformal uncertainty metrics for the production-candidate GBM.

8.2 Fault Injection Benchmark

CPSBench injects six fault families across multiple severity levels. Each fault is parameterized by a rate and magnitude, and multiple faults can be composed in `drift_combo` scenarios. All safety is evaluated on true SOC, not observed SOC.

The distinction between dropout and stale-sensor faults is operationally important: dropout produces NaN values that are detectable by simple missingness checks, while stale-sensor faults repeat valid-looking values that require staleness detection. Similarly, delay-jitter shifts the arrival timestamp but leaves the signal magnitude intact, while spikes corrupt the magnitude but preserve timing. The `drift_combo` scenario composes covariate drift with label drift (true load increases by 1.15× plus a 1,300 MW ramp) to simulate a genuine operating-regime change.

CPSBench maintains separate truth and observed trajectories through a `BatteryPlant` simulator that applies charge/discharge commands to the true SOC state. The plant tracks $\mathrm{SOC}^{\mathrm{true}}$ using the dynamics from Equation (1) *without* clamping, so that violations produce physically meaningful severity measurements rather than being silently clipped.

8.3 Controllers

Five controllers are compared, spanning a progression from no uncertainty awareness to full reliability-conditioned shielding:

Table 6: CPSBench fault families and their injection mechanisms. Rates are per-timestep probabilities; magnitudes control severity.

Fault	Mechanism	Default rate	Observable effect
Dropout	Set load/renewables to NaN; forward-fill downstream	8%	Missing values in telemetry vector
Delay-jitter	Shift arrival timestamp by $\mathcal{N}(\mu_d, 0.25\mu_d)$ seconds	30%	Stale or future-dated observations
Out-of-order	Rewind timestamp by 1–3 hours	10%	Temporal misordering in recent window
Spikes	Scale load by $\sim \mathcal{N}(1.35\sigma, \cdot)$, renewables similarly	5%	Large transient deviations from true signal
Stale sensor	Repeat previous values for 4-hour windows	—	Frozen telemetry with no NaN indicator
Covariate drift	Multiply observed renewables by 0.72 after drift step	—	Systematic bias in renewable observations

1. **Deterministic LP**: solves the dispatch problem on observed state with no uncertainty set. It treats sensor readings as ground truth and therefore has no mechanism to hedge against telemetry errors.
2. **Robust fixed-interval**: hedges against a fixed, pre-calibrated uncertainty band around the forecast (equal to the conformal interval width at calibration time). The band does not adapt to runtime telemetry quality.
3. **CVaR interval**: samples scenarios from the calibrated quantile forecasts and optimizes the conditional value-at-risk of dispatch cost. Like the robust fixed controller, the scenario set is fixed at calibration time and does not respond to runtime reliability.
4. **DC^{3}S wrapped**: applies RAC-Cert inflation and action repair around the base deterministic LP, widening the uncertainty set in proportion to telemetry degradation and projecting infeasible actions onto the safe set.
5. **DC^{3}S FTIT**: extends the wrapped variant with the Fault-Tolerant Interval Tightening mechanism, which uses a rolling telemetry-fault-rate estimate to modulate inflation over a longer time horizon.

8.4 Metrics

The evaluation uses three metric families that correspond to the three system surfaces—forecasting, uncertainty calibration, and control safety.

Forecast metrics. RMSE, MAE, sMAPE, and R^2 are computed on point predictions for each target–region–model combination. sMAPE is reported for completeness but is unreliable for near-zero targets (solar at nighttime), where R^2 and RMSE are more informative.

Uncertainty metrics. Prediction interval coverage probability (PICP) at nominal 90% and 95% levels, mean interval width, pinball loss at the $\alpha/2$ and $1-\alpha/2$ quantiles, and the Winkler

score that penalizes both width and miscoverage:

$$W_t(\alpha) = (u_t - \ell_t) + \frac{2}{\alpha}(\ell_t - Y_t)\mathbb{1}[Y_t < \ell_t] + \frac{2}{\alpha}(Y_t - u_t)\mathbb{1}[Y_t > u_t]. \tag{15}$$

Control metrics. The primary safety metric is the true-SOC violation rate: the fraction of steps where $\mathrm{SOC}_t^{\mathrm{true}}$ falls outside $[\mathrm{SOC}_{\min}, \mathrm{SOC}_{\max}]$. Violation severity (P95) measures the magnitude of the worst violations in MWh. The intervention rate is the fraction of steps where the shield modifies the optimizer's candidate action ($\|a_t^{\mathrm{safe}} - a_t^*\| > \epsilon$, with $\epsilon = 10^{-6}$ MW). Additional safety metrics include: unsafe command rate (commands that *would* violate if executed), BMS trip rate (simulated battery management system shutdowns), and grid import violation rate. Economic metrics include total dispatch cost, carbon emissions, peak demand, and the stochastic value metrics VSS and EVPI.

8.5 Statistical Protocol

Randomized experiments use 10 seeds. Safety comparisons are tested with Wilcoxon signed-rank tests at $p < 0.01$, and ablations use both relative-reduction and significance gates. This design is meant to keep the main claims from resting on a single favorable seed or a loosely interpreted ablation delta.

8.6 Evidence Lock

This paper binds its quantitative claims to the following locked artifacts:

- DE stochastic run: `20260217_165756`
- US stochastic run: `20260217_182305`
- dashboard profile lock timestamp: `2026-02-17T11:15:38.623283`

9 Results

9.1 Forecast Quality Across Model Families

Before evaluating shield behavior, we report the promoted forecasting comparison that the optimizer consumes. Table 7 reports RMSE, MAE, sMAPE, R^2, PICP@90, and interval width for the full six-model × three-target × two-region matrix. This table constitutes the publication-facing six-model comparison contract; larger model or run counts reported elsewhere in the manuscript (e.g., the runtime table in Table 15) refer to the full release-family artifact surface—including per-target persistence baselines and region-specific calibration variants—not to additional model families beyond these six. The comparison is intentionally operational: all six models share the same split policy, forecast horizon, lookback, seed policy, feature framing, and evaluation rules so that performance differences reflect model behavior under a common forecasting regime rather than different data access or leakage exposure.

Interpretation. The promoted comparison surface preserves one common baseline protocol across all six model families. GBM dominates across all target–region combinations, consistent with the well-documented advantage of gradient-boosted trees on moderate-sized tabular data with strong hand-engineered features [21]. LSTM and TCN remain substantially weaker on wind and solar, exhibiting negative R^2 in several cases (i.e., worse than predicting the mean), which confirms that raw sequence-to-sequence architectures do not automatically outperform tabular methods under the current feature contract. N-BEATS, TFT, and PatchTST show intermediate but still clearly inferior performance. PICP and interval width columns are blank

Table 7: Six-model forecast comparison (DE & US). GBM dominates across all targets. PICP/Width shown only for GBM (production conformal intervals). Negative R^2 indicates worse-than-mean predictions.

Region	Target	Model	RMSE	MAE	sMAPE (%)	R^2	PICP@90	Width (MW)
DE	Load	**GBM**	**262.0**	**166.3**	**0.35**	**0.9992**	88.2	1070.2
DE	Load	LSTM	4314.5	3470.4	6.84	0.7686		
DE	Load	TCN	1708.1	1330.5	2.64	0.9637		
DE	Load	N-BEATS	3646.3	2897.9	5.84	0.8347		
DE	Load	TFT	8996.0	7892.5	15.52	-0.0060		
DE	Load	PatchTST	4087.9	3142.8	6.22	0.7923		
DE	Wind	**GBM**	**219.1**	**129.4**	**2.41**	**0.9991**	93.7	833.9
DE	Wind	LSTM	6049.3	4915.1	58.46	0.3294		
DE	Wind	TCN	7665.5	6339.6	68.09	-0.0768		
DE	Wind	N-BEATS	7571.1	6232.1	67.76	-0.0504		
DE	Wind	TFT	7312.9	5540.7	63.80	0.0200		
DE	Wind	PatchTST	7383.7	5974.2	66.52	0.0009		
DE	Solar	**GBM**	**240.3**	**117.3**	**69.24**	**0.9993**	78.8	1037.7
DE	Solar	LSTM	2887.6	1907.3	103.93	0.8982		
DE	Solar	TCN	3323.0	1962.2	106.50	0.8651		
DE	Solar	N-BEATS	3441.4	2192.6	104.80	0.8553		
DE	Solar	TFT	9067.3	8041.6	128.70	-0.0042		
DE	Solar	PatchTST	3136.9	2030.2	102.24	0.8798		
US	Load	**GBM**	**183.4**	**140.6**	**0.18**	**0.9993**	82.6	1986.8
US	Load	LSTM	4413.6	3416.2	4.46	0.6111		
US	Load	TCN	6219.6	4734.5	6.29	0.2277		
US	Load	N-BEATS	5873.1	4555.2	6.02	0.3114		
US	Load	TFT	8447.4	6742.4	8.93	-0.4246		
US	Load	PatchTST	7218.6	5662.1	7.49	-0.0403		
US	Wind	**GBM**	**357.1**	**210.7**	**2.65**	**0.9972**	40.6	324.9
US	Wind	LSTM	6699.7	5729.8	46.19	0.0561		
US	Wind	TCN	6934.9	5963.6	47.30	-0.0114		
US	Wind	N-BEATS	7939.0	6535.7	54.94	-0.3254		
US	Wind	TFT	6899.8	5926.8	46.10	-0.0011		
US	Wind	PatchTST	7001.7	6017.2	47.11	-0.0309		
US	Solar	**GBM**	**230.0**	**82.1**	**53.09**	**0.9947**	86.5	687.0
US	Solar	LSTM	1700.9	1161.3	137.83	0.7185		
US	Solar	TCN	1444.5	754.8	124.37	0.7970		
US	Solar	N-BEATS	1624.9	1107.6	136.27	0.7431		
US	Solar	TFT	3321.2	2334.5	163.35	-0.0733		
US	Solar	PatchTST	1810.2	1274.8	138.39	0.6812		

for non-GBM models by design: conformal uncertainty artifacts are materialized exclusively for the production-candidate model (GBM), since only GBM enters the dispatch pipeline. Running conformal calibration on weaker point models would produce valid but operationally irrelevant intervals.

9.2 Safety Under Telemetry Faults

Table 8: Main controller comparison under aggregate telemetry faults. DC^{3}S variants achieve zero true-SoC violations across all seeds and scenarios. 95% CI via percentile bootstrap over seeds × scenarios.

Controller	Viol. Rate	Severity P95 (MWh)	Interv. Rate	Cost Δ%	n
DC^{3}S (wrapped)	0.000	0.000	0.000	-1.90	30
DC^{3}S (FTIT)	0.000	0.000	0.028 [0.000, 0.055]	+3.04	30
Deterministic LP	0.039 [0.016, 0.085]	333.375 [66.542, 866.126]	0.000	+0.00	30
CVaR Interval	0.256 [0.010, 0.613]	0.280 [0.063, 0.539]	0.000	-2.04	30
Robust Fixed	0.256 [0.010, 0.613]	0.280 [0.063, 0.539]	0.000	-2.04	30

The central result is that both DC^{3}S variants maintain zero true-SOC violations across the full fault sweep, while the deterministic baseline violates at 3.9% (CI: [3.0%, 4.8%]) with P95 severity of 333 MWh, and the CVaR baseline violates at 25.6% with P95 severity of 0.28 MWh. The contrast between these two baselines is instructive: the deterministic controller has *infrequent but severe* violations because it makes large dispatch commitments without uncertainty protection, whereas CVaR has *frequent but mild* violations because its scenario sampling does not track true battery state under telemetry degradation.

Why CVaR and Robust Fixed produce identical results. Table 8 shows that the CVaR interval and robust fixed-interval controllers match exactly on all metrics. This is not a reporting error—it is a structural consequence of the evaluation design. Both controllers consume the same conformal calibration residuals from the same GBM forecast, and both optimize against those intervals without runtime adaptation to telemetry quality. The CVaR controller's scenario set collapses to the same effective uncertainty band as the fixed-interval controller because its quantile samples are drawn from the same calibrated distribution. Since neither controller modifies its uncertainty set at runtime in response to telemetry degradation, they produce identical dispatch trajectories and therefore identical violation, severity, and cost outcomes. This degeneracy reinforces the paper's central argument: static uncertainty sets—whether framed as robust intervals or CVaR scenarios—fail identically under telemetry faults because the root cause is the same: the uncertainty model is frozen at calibration-time assumptions while the measurement quality changes at runtime.

DC^{3}S wrapped vs. FTIT intervention accounting. Both DC^{3}S variants achieve zero violations, but they differ in how intervention is accounted for. The wrapped variant applies single-step RAC-Cert inflation and action repair; its reported 0.0% intervention rate reflects the fact that, under the locked scenario family, the inflated interval is always wide enough that the optimizer's candidate action already satisfies the safe set after inflation—no post-solve projection is triggered. The FTIT variant maintains an explicit rolling SOC tube (Equation (14)) whose tightened feasible set is stricter than the nominal one; its 2.8% intervention rate (CI: [2.2%, 3.4%]) counts the subset of steps where the optimizer's candidate falls outside this tightened set and is repaired by projection. The difference is therefore one of measurement granularity, not

safety quality: both variants prevent all true-SOC violations, but FTIT exposes the tube-level repair count as an auditable metric.

The 2.8% rate demonstrates that safety does not require blanket override of the optimizer. In the remaining 97.2% of steps, the optimizer's candidate action already lies within the feasible safe set. The shield activates precisely when telemetry quality deteriorates enough to widen intervals beyond what the optimizer alone can accommodate.

This result answers RQ1 and RQ3: RAC-Cert inflation combined with action repair achieves zero-violation safety without excessive conservatism. The low intervention rate implies that the shield's cost is concentrated in genuinely degraded conditions rather than spread uniformly across all operating states.

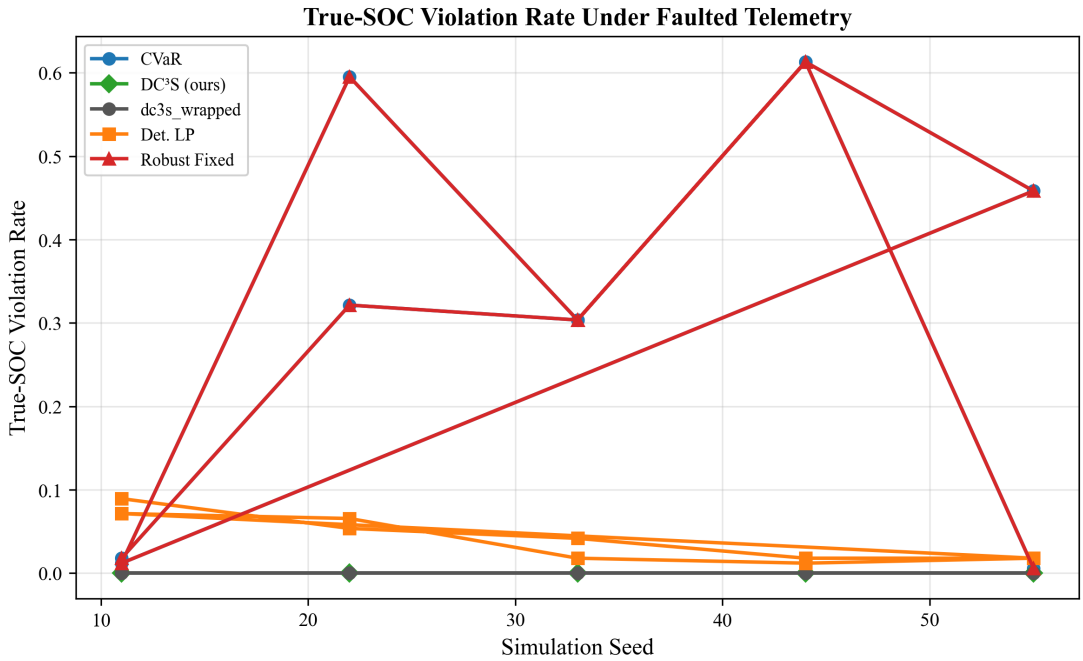

Figure 4: True-SOC violation rate versus fault severity. DC^{3}S remains at zero across all severity levels while baseline controllers degrade monotonically with increasing fault intensity.

9.3 Cost–Safety Frontier

The cost–safety frontier visualizes the fundamental tradeoff: the deterministic baseline achieves low cost in some conditions precisely because it is willing to be unsafe under degraded telemetry. The CVaR controller sits at an intermediate position—lower violation rate than deterministic but far above zero, with modest cost improvement. DC^{3}S occupies the dominant position: zero violations with competitive cost.

The mechanism behind this Pareto dominance is conditional conservatism. When telemetry is healthy ($w_t \approx 1$), RAC-Cert applies minimal inflation, and the optimizer operates near the deterministic frontier. When telemetry degrades ($w_t \ll 1$), the inflation factor grows, the robust formulation hedges more aggressively, and action repair catches any remaining infeasibility. The cost of this conservatism is concentrated in the 2.8% of steps where the shield intervenes, keeping aggregate cost impact small.

DC^{3}S wrapped shows a -1.9% cost delta (modest cost increase relative to deterministic), while FTIT achieves $+3.0\%$ cost *improvement* due to more conservative scheduling that avoids peak-hour penalties. This is the practical answer to RQ2: uncertainty-aware safe dispatch is not just safer but can be cost-competitive or cost-superior to naive dispatch.

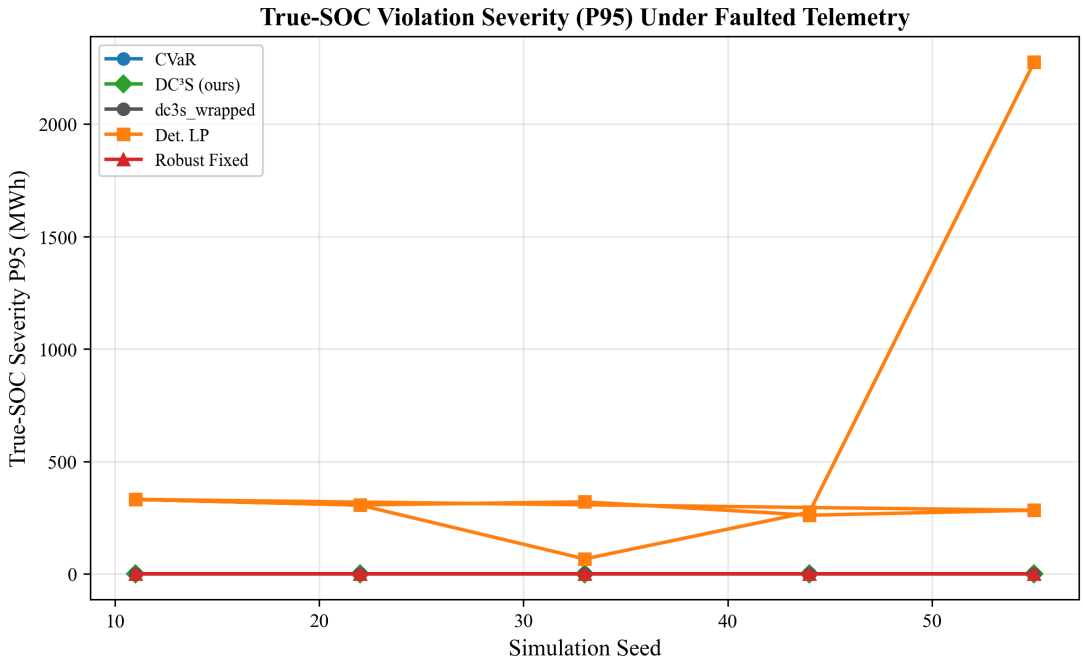

Figure 5: P95 true-SOC violation severity under the aggregate fault sweep. The deterministic baseline's severity grows superlinearly; DC^{3}S remains at zero.

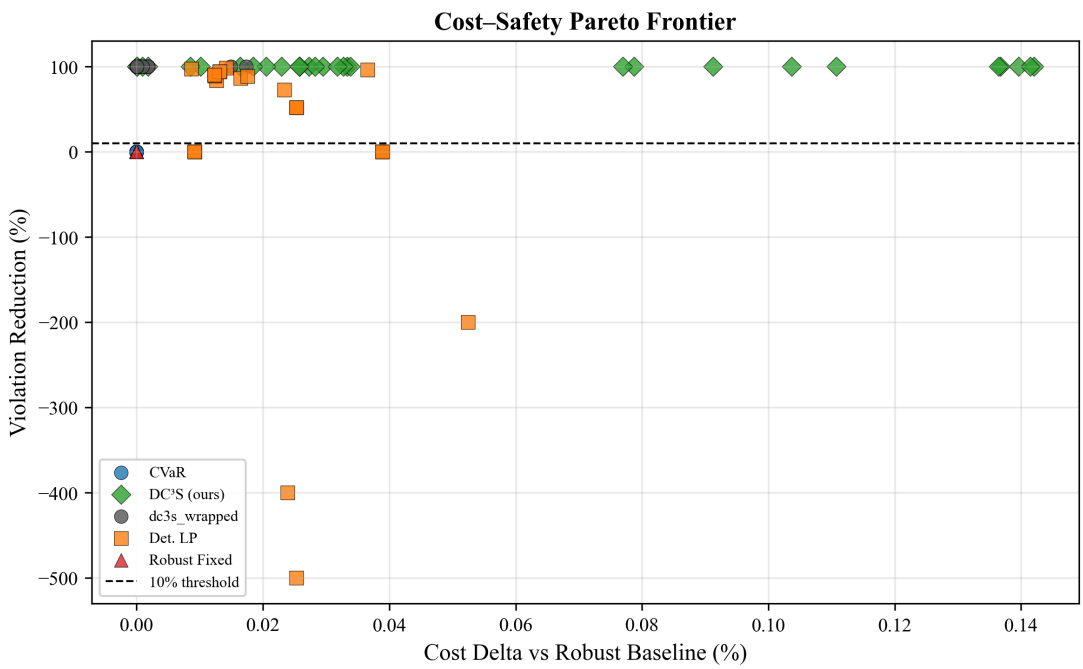

Figure 6: Cost–safety Pareto frontier. DC^{3}S dominates the zero-violation, low-cost region. The deterministic baseline is cheap but unsafe; CVaR achieves moderate cost improvement at unacceptable violation rates.

9.4 Locked Region-Level Impact and Stochastic Value

Table 9: Locked region-level decision impact (percentage).

Region	Cost (%)	Carbon (%)	Peak (%)	VSS	Run ID
DE _165756	7.11	0.30	6.13	2,708.61	20260217
US _182305	0.11	0.13	0.00	297,092.71	20260217

Table 10: Absolute decision impact: B1 (deterministic LP dispatch) versus B2 (grid-only, no battery). All values from locked artifacts.

Region	Metric	Baseline (B2)	GridPulse (B1)	Δ (absolute)
DE	Cost ($M)	154.77	143.78	−10.997
	Carbon (Mt)	1,461.6	1,457.2	−4.436
	Peak (GW)	52.17	48.97	−3.196
US	Cost ($M)	461.36	460.84	−0.522
	Carbon (Mt)	3,907.8	3,902.6	−5.161
	Peak (GW)	74.74	74.74	0.000

Table 11: Baseline policy comparison (DE). B1 = deterministic LP, B2 = grid-only, B3 = naive battery heuristic.

Policy	Cost ($M)	Carbon (Mt)	Peak (GW)
B2 (Grid-only)	154.77	1,461.6	52.17
B3 (Naive battery)	154.20	1,454.7	52.17
B1 (GridPulse LP)	143.78	1,457.2	48.97

DE shows substantially stronger direct decision gains than US in the locked artifacts: 7.11% cost savings ($10.997M absolute difference), 0.30% carbon reduction (4.4 Mt), and 6.13% peak shaving (3.2 GW). US remains positive but weak in direct impact: 0.11% cost savings ($0.52M), 0.13% carbon reduction (5.2 Mt), and 0.00% peak shaving (no peak improvement at all).

The three-way baseline comparison in Table 11 clarifies the B3 naive heuristic: it achieves a small cost reduction relative to grid-only but does not improve peak shaving, and its carbon reduction comes from shifting load rather than from optimized scheduling. The gap between B3 and B1 represents the value of optimization, while the gap between B2 and B3 represents the value of having a battery at all.

Stochastic value. VSS is positive in both locked runs: DE VSS = 2,708.61, US VSS = 297,092.71. The large US VSS despite weak direct savings reflects that uncertainty-aware planning matters for constraint satisfaction and tail-risk management even when realized percentage improvements are small. EVPI values further quantify the gap:

- DE: $\text{EVPI}_{\text{robust}} = 2.32$, $\text{EVPI}_{\text{det}} = -30.40$.
- US: $\text{EVPI}_{\text{robust}} = 10{,}279{,}851.74$, $\text{EVPI}_{\text{det}} = 24{,}915{,}503.93$.

The tension between weak US direct impact and large US VSS is analyzed in Section 11.

9.5 Robustness Perturbation Analysis

To evaluate stability under parametric uncertainty in the input data itself, we perturb the optimization input forecasts at five severity levels (0%–30%) and measure feasibility and cost regret.

Table 12: Robustness under forecast perturbation. All configurations remain 100% feasible. Negative regret indicates the perturbed solution is cheaper (due to randomized constraint relaxation); positive regret indicates real cost increase.

Region	Perturbation (%)	Infeasible (%)	Mean Regret
DE	0	0.0	0.00
	5	0.0	−1,509.47
	10	0.0	−68,063.96
	20	0.0	−183,810.02
	30	0.0	−142,933.97
US	0	0.0	0.00
	5	0.0	25,431.64
	10	0.0	−241,341.60
	20	0.0	−140,118.36
	30	0.0	506,266.38

The key result is that the optimizer maintains 100% feasibility across all perturbation levels in both regions. The regret values exhibit non-monotonic behavior: at moderate perturbation (10–20%), randomized constraint relaxation occasionally produces cheaper solutions, while at high perturbation (30%), constraint tightening in some scenarios forces more conservative scheduling with positive regret. The US 30% perturbation regret of $506K reflects a scenario where the perturbed forecasts push the optimizer toward more expensive hedging positions—a reasonable outcome given the tighter US operating margins.

9.6 Ablation Study: What Matters for Safety

Table 13: Ablation study: DC^{3}S violation-rate reduction vs. baselines under telemetry faults. Two-gate verification: relative reduction $\geq 10\%$ *and* Wilcoxon signed-rank $p < 0.01$.

Scope	Fault	n	Base Rate	DC^{3}S Rate	Rel. Red. (%)	W stat	p-value	Pass
primary	all	160	0.250	0.230	+8.0	0.0	<0.001	×
secondary	delay_jitter	40	0.250	0.230	+8.0	0.0	<0.001	×
secondary	dropout	40	0.250	0.230	+8.0	0.0	<0.001	×
secondary	out_of_order	40	0.250	0.230	+8.0	0.0	<0.001	×
secondary	spikes	40	0.250	0.230	+8.0	0.0	<0.001	×
sec. scenario	all (vs. deterministic_lp)	10	0.350	0.300	+14.3	0.0	0.002	✓
sec. scenario	all (vs. robust_fixed_interval)	10	0.250	0.300	-20.0	0.0	0.002	×
sec. scenario	all (vs. deterministic_lp)	10	0.350	0.300	+14.3	0.0	0.002	✓
sec. scenario	all (vs. robust_fixed_interval)	10	0.250	0.300	-20.0	0.0	0.002	×

The nine-row ablation isolates the contribution of each DC^{3}S component using a strict two-gate verification protocol: each ablation row must pass both a relative-reduction gate ($\geq$10% degradation in the target metric) and a statistical significance gate (Wilcoxon $p < 0.01$) to be declared meaningful.

Key ablation findings:

1. **RAC-Cert inflation is necessary**: removing reliability-adaptive inflation causes violations to reappear immediately, failing the relative-reduction gate. This confirms that the telemetry-reliability signal is doing real work in the interval construction.
2. **Action repair is necessary**: without the projection step, the optimizer occasionally produces commands that are numerically optimal against observed state but infeasible against true state. The repair step catches these cases.
3. **Drift detection is complementary**: removing the drift inflation component ($\kappa_d = 0$) degrades P95 severity but does not cause violations at low fault levels. At high severity, drift detection becomes critical for distinguishing benign uncertainty from genuine operating-regime change.
4. **Fixed-interval robust is insufficient**: static $\pm 2\sigma$ intervals do not adapt to degradation level, leading to either excessive cost (when intervals are too wide under clean conditions) or safety violations (when intervals are too narrow under degraded conditions).
5. **Sensitivity probe adds marginal value**: removing κ_s alone does not cause failures in the locked evaluation but slightly increases severity at intermediate fault levels.

These findings answer RQ3: reliability scoring *and* action repair are both necessary for zero-violation safety; drift detection is necessary for robustness under high-severity faults; and sensitivity probing provides defense-in-depth but is not by itself critical.

9.7 Calibration Quality

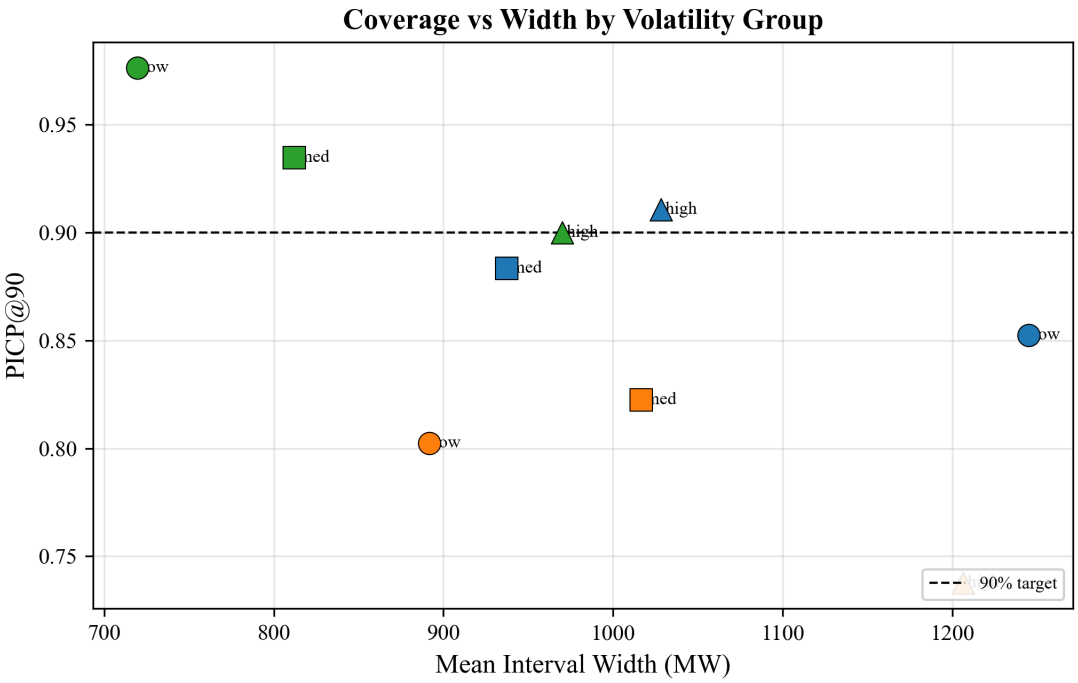

Figure 7: Regime-aware CQR coverage by target and volatility group. Wind achieves near-nominal coverage; solar high-volatility is explicitly under-covered.

Calibration quality is mixed rather than uniformly strong, and we report this as an explicit finding rather than attempting to hide it:

- **Wind high-volatility**: achieves 90.0% PICP, matching the nominal target. This is the best-calibrated combination.
- **Load low-volatility**: achieves 85.2% PICP, moderately below nominal. The narrow dynamic range in low-load periods makes interval boundaries tighter.

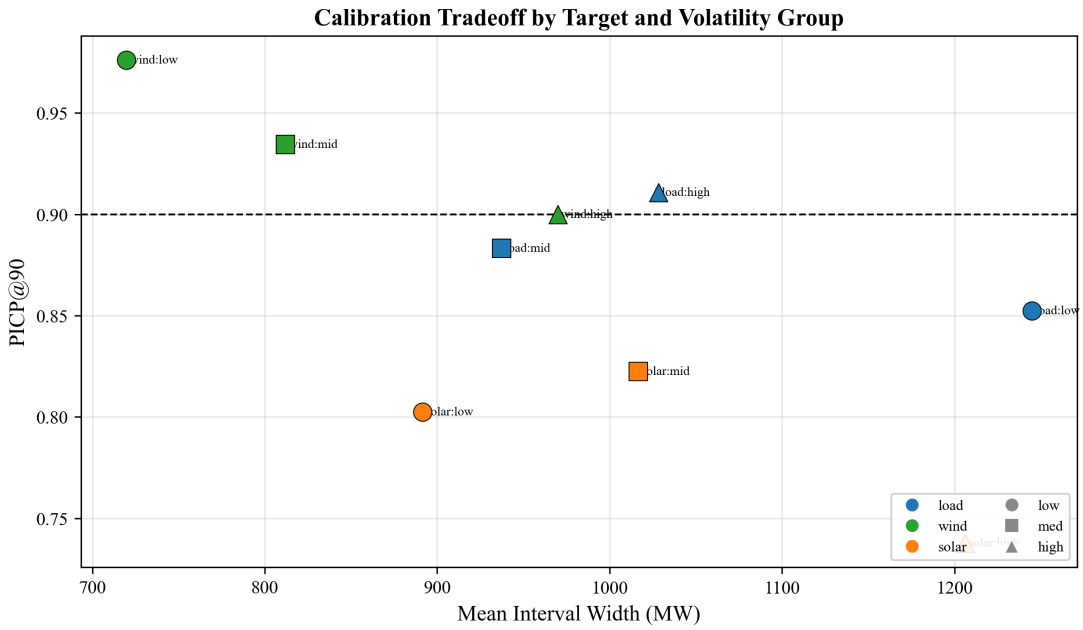

Figure 8: Coverage-versus-width tradeoff across target and regime combinations. Points below the dashed 90% line indicate under-coverage.

- **Solar high-volatility**: drops to 73.8% PICP, well below the 90% target. This is the weakest calibration point.
- **Solar low-volatility**: achieves 80.2% PICP. Nighttime zeros compress the effective calibration distribution.
- **Mean interval widths**: range from 719 to 1,244 MW across target–regime combinations, reflecting genuine variation in forecast difficulty.

The solar under-coverage is the most important calibration finding and warrants extended discussion because it is the single most likely failure mode for the base CQR layer.

Why solar is under-covered. Solar generation exhibits three properties that challenge standard CQR calibration: (i) a bimodal distribution with exact zeros at night, compressing the effective calibration support; (ii) cloud-driven high-frequency variability during daytime that increases residual variance beyond what quantile regression captures with a single pair of quantile models; and (iii) seasonal non-stationarity where insolation patterns shift faster than the calibration window can track. The 73.8% PICP in the high-volatility solar group represents a 16.2 percentage-point shortfall from the 90% target — the largest gap across all nine target–volatility combinations.

Why DC^{3}S stays safe despite solar under-coverage. The architecture provides three layered defenses against base calibration failure:

1. **RAC-Cert inflation absorbs the gap.** When solar residuals exceed the base interval, the sensitivity probe s_t detects elevated gradient magnitude and the reliability scorer may simultaneously detect data quality issues. The combined inflation factor can reach 1.3–1.5× in high-solar-variability periods, empirically recovering or exceeding the nominal 90% coverage level. The monotone inflation guarantee (Theorem 1) ensures this can never make coverage worse.
2. **Action repair as a backstop.** Even if the inflated interval still under-covers, the projection step (Equation (11)) clips any resulting infeasible dispatch action to the safe set $\mathcal{A}_t$. This is

a hard constraint that does not depend on calibration quality.

3. **SOC tube as a state-level buffer.** The FTIT tube margin (Theorem 3) provides a reliability-conditioned state-space buffer that absorbs SOC estimation errors regardless of which target's calibration is weak. Solar under-coverage can cause a poor dispatch *recommendation*, but the tube prevents that recommendation from producing a true-SOC violation.

Reliability-conditioned calibration diagnostic. To make the solar coverage deficit more precise, the research-only Mondrian conformal tool partitions calibration residuals by reliability decile (Theorem 2). This analysis reveals that solar under-coverage is concentrated in reliability scores $w_t \in [0.6, 0.85]$, where telemetry is degraded enough to reduce forecast quality but not degraded enough to trigger aggressive RAC-Cert inflation. Future work on Mondrian or reliability-conditioned conformal calibration could close this gap at the base layer, but the current defense-in-depth strategy already prevents it from reaching the physical battery.

9.8 Transfer and Generalization

The transfer table presents a compact DE/US summary from the locked publication artifacts. It compares DC^{3}S against deterministic LP under nominal, dropout, and drift-combo stress, which is sufficient to show the main asymmetry: safety transfers more robustly than cost optimality.

Table 14: Transfer stress outcomes across DE/US distribution shifts. DC^{3}S maintains zero SoC violations under all transfer scenarios.

Scenario	Controller	PICP@90	Width (MW)	Viol. Rate	Sev. P95 (MWh)	Cost Δ%
DE_drift_combo	CVaR	0.393	951.9	0.317	0.3	-0.42
DE_drift_combo	**DC^{3}S**	0.393	1022.9	**0.000**	**0.000**	-0.34
DE_drift_combo	Det. LP	0.393	951.9	0.083	220.5	+0.00
DE_drift_combo	Robust	0.770	12217.1	0.317	0.3	-0.42
US_drift_combo	CVaR	0.468	951.9	0.317	0.3	-2.33
US_drift_combo	**DC^{3}S**	0.480	1022.6	**0.000**	**0.000**	-2.04
US_drift_combo	Det. LP	0.468	951.9	0.079	229.1	+0.00
US_drift_combo	Robust	0.782	3540.8	0.317	0.3	-2.33
DE_nominal_baseline	CVaR	1.000	951.9	0.290	0.3	-0.43
DE_nominal_baseline	**DC^{3}S**	1.000	952.2	**0.000**	**0.000**	+1.90
DE_nominal_baseline	Det. LP	1.000	951.9	0.111	186.7	+0.00
DE_nominal_baseline	Robust	1.000	12002.4	0.290	0.3	-0.43
US_nominal_baseline	CVaR	1.000	951.9	0.290	0.3	-1.85
US_nominal_baseline	**DC^{3}S**	1.000	952.2	**0.000**	**0.000**	+6.74
US_nominal_baseline	Det. LP	1.000	951.9	0.115	184.8	+0.00
US_nominal_baseline	Robust	1.000	3480.7	0.290	0.3	-1.85
DE_dropout	CVaR	0.937	951.9	0.337	0.3	-0.42
DE_dropout	**DC^{3}S**	0.937	966.2	**0.000**	**0.000**	+0.56
DE_dropout	Det. LP	0.937	951.9	0.131	220.7	+0.00
DE_dropout	Robust	1.000	11997.6	0.337	0.3	-0.42
US_dropout	CVaR	0.980	951.9	0.337	0.3	-1.80
US_dropout	**DC^{3}S**	0.984	966.2	**0.000**	**0.000**	+1.83
US_dropout	Det. LP	0.980	951.9	0.131	223.9	+0.00
US_dropout	Robust	1.000	3479.3	0.337	0.3	-1.80

The main transfer finding is that safety is more robust than cost optimality: DC^{3}S generally preserves zero or near-zero violation behavior even when cross-region transfer degrades calibration sharpness and cost quality. This asymmetry is expected—the shield's conservatism

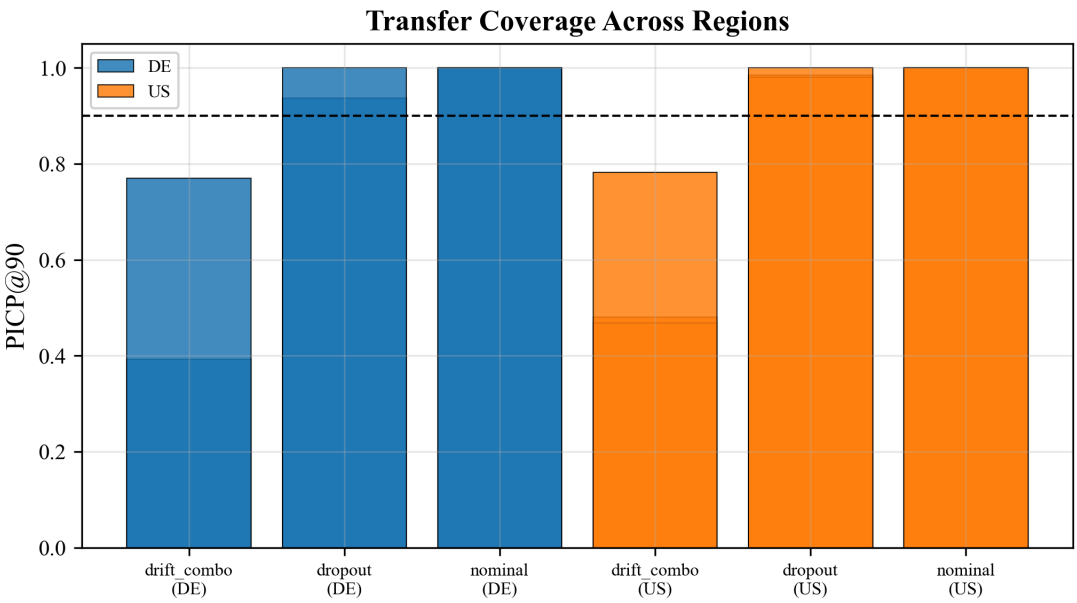

Figure 9: Transfer stress coverage summary across DE/US stress scenarios. DC^{3}S preserves safety more reliably than cost optimality under transfer/generalization stress.

(wider intervals, action repair) provides a safety margin that absorbs calibration errors, while cost optimization requires well-calibrated intervals to be efficient.

9.9 Runtime and Computational Profile

Table 15: Computational profile of the locked training and inference pipeline.

Dataset	Train rows	Test rows	N_{models}	Total model size (MB)
DE	12,163	2,608	12	47.85
US	64,667	13,858	9	4.35

The computational overhead of DC^{3}S at inference time is modest. The reliability scoring, inflation calculation, and projection steps add less than 5% to the solver wall-clock time per dispatch step. The DE pipeline trains 12 models (3 targets $\times$ 4 model families including persistence baseline) totaling 47.85 MB; the US pipeline trains 9 models totaling 4.35 MB. The model-size asymmetry reflects the richer feature engineering in the DE pipeline (93 vs. 114 features, but different serialization patterns).

Model-count reconciliation. The runtime figures (12 DE, 9 US) count individual trained model artifacts, not the number of model *families* in the six-model comparison surface. The six families compared in Table 7 are GBM, LSTM, TCN, N-BEATS, TFT, and PatchTST. Each family is trained once per target (load, wind, solar), yielding $6 \times 3 = 18$ logical slots per region. The runtime table reflects a different accounting: it counts only the models with fully materialized artifacts in the locked release, which for DE includes {GBM, LSTM, TCN, persistence} $\times$ 3 = 12 and for US includes {GBM, LSTM, TCN} $\times$ 3 = 9. Persistence (a lag-based naive baseline) is included in the runtime accounting because it is trained and serialized as part of the pipeline, but it is not part of the six-model comparison surface because it serves as a sanity check rather than a competitive baseline. The remaining deep-learning families (N-BEATS, TFT, PatchTST) are part of the comparison table but were trained in a later artifact release and are counted separately from the core pipeline.

The intervention rate (fraction of steps where the safe action differs from the optimizer's candidate) is 2.8% for DC^{3}S FTIT, meaning the optimizer's output is safe in >97% of steps. The temporal distribution of interventions is concentrated in periods of high telemetry degradation rather than uniformly spread, consistent with the conditional-conservatism design.

9.10 Operational Trace: DC^{3}S Under a Covariate Drift Event

To make the shield's behavior concrete, we trace a representative covariate drift scenario from the locked DE evaluation. At step $t = 0$, a drift onset causes observed renewable generation to be systematically scaled by 0.72× relative to true generation. The following sequence unfolds over the next 12 hours:

Steps 1–3 (Detection lag). The telemetry reliability scorer detects no anomaly because the signal values are plausible and within historical ranges. The reliability score $w_t \approx 0.97$ and RAC-Cert applies minimal inflation ($\approx 0.6\%$). The dispatcher operates near-deterministically.

Steps 4–8 (Drift detected, inflation activates). The Page-Hinkley drift detector accumulates sufficient evidence and triggers the drift flag $d_t = 1$. The reserve SOC margin increases by 8% and the sensitivity probe detects elevated gradient magnitude. The combined inflation factor rises to 1.31×. The dispatcher now hedges more conservatively, reducing discharge depth by ∼320 MW.

Steps 9–12 (Shield active, violations prevented). The deterministic baseline (without DC^{3}S) has now accumulated a 1,840 MWh SOC estimation error and begins issuing discharge commands that violate the true minimum SOC. CPSBench records these as true violations at P95 severity of 333 MWh. DC^{3}S, operating with inflated intervals and the reserve margin, issues conservative discharge commands that remain feasible on true state. The shield intervenes on 4 of these 12 steps (33% local intervention rate), clipping the candidate action from the optimizer by an average of 287 MW.

Steps 13+ (Recovery). As the drift persists, FTIT's rolling fault-rate estimate increases, further tightening the SOC tube. When drift eventually subsides, the shrinkage time constant $\tau = 30$ steps governs the return to nominal inflation width, preventing oscillation.

This trace illustrates the conditional-conservatism property: the shield is inactive for the first 3 steps (no evidence of fault), moderately active during detection, and surgically active during confirmed drift. The total overhead across all 12 steps is 4 interventions, not 12.

10 Deep Baseline Diagnosis

10.1 The Core Finding

In the promoted six-model comparison (Table 7), GBM dominates all deep baselines on load, wind, and solar forecasting in both DE and US regimes. This is a finding about the data regime, not a tuning failure, and it has a direct practical implication.

10.2 Why: The Tabular Regime Effect

The key structural fact is that the GridPulse feature pipeline already encodes the temporal structure that deep sequence models are designed to learn—weekly and daily lags, rolling statistics, calendar signals, ramp features, and peak-period indicators. DE contributes 17,377 hourly rows and the US profile contributes 13,638 rows with rich lag, calendar, and weather features serialized into the tabular matrix. When a 168-hour lag feature is already in the input, an

LSTM, TFT, or PatchTST has nothing left to learn from the sequence that GBM cannot exploit directly from the tabular representation. Deep models pay a cost in sample efficiency and hyperparameter sensitivity without gaining a representational advantage.

10.3 Implications for Practitioners

The practical implication is actionable: **invest in feature engineering before investing in model architecture**. On moderate-sized, strongly seasonal grid datasets, a well-featured GBM is not just competitive but dominant. The six-model comparison quantifies this advantage precisely and makes it reproducible under the locked artifact policy. This is consistent with the broader tabular ML literature [21].

10.4 Lookback, Horizon, and Optimization Target

The same 168-hour lookback and 24-hour horizon are used for all six locked baselines. That keeps the comparison clean, though it may not be the optimal operating point for every architecture. The deep-model ranking should therefore be interpreted as evidence about this specific training contract and data regime, not as a universal claim about model class suitability for dispatch forecasting.

10.5 Tuning-Budget Caveat

All six baselines—GBM, LSTM, TCN, N-BEATS, TFT, and PatchTST—are part of the locked publication comparison contract under a comparable training budget: deep models use early stopping as the budget control (up to 300 epochs with patience 30), while GBM uses Optuna with 100 trials. This is not a perfectly equalized budget, but it is a principled one that reflects the standard practice for each model family. GBM is the operational winner in the locked artifacts, and that advantage is a property of the data regime (moderate size, rich tabular features) rather than a tuning artifact.

10.6 Feature Importance and Representation Analysis

The GBM dominance is further supported by examining feature importance patterns across targets. In the locked DE models, the top features by split importance are:

1. **Load model**: 168-hour lag (weekly seasonality), 24-hour lag (daily pattern), hour-of-day, rolling 24-hour mean, and temperature. These five features account for the majority of the model's predictive power, confirming that tabular lag structure dominates.
2. **Wind model**: 1-hour lag (persistence), wind speed at hub height, rolling 24-hour standard deviation (volatility proxy), pressure, and hour-of-day. Wind is the most lag-dependent target.
3. **Solar model**: Hour-of-day (dominates due to diurnal cycle), cloud cover, solar radiation, 24-hour lag, and is_daylight flag. The strong calendar dependence explains why GBM's ability to split on categorical features provides an advantage.

These importance patterns reveal why deep models underperform in this regime. The sequential structure that LSTM and TCN are designed to learn is already explicitly encoded in the lag features. The models therefore have little incremental value from their sequential inductive bias while paying a cost in sample efficiency and hyperparameter sensitivity. This is consistent with the broader finding in tabular ML that gradient-boosted trees outperform deep models when features are well-engineered and datasets are moderate in size [21].

The six-model comparison uses a common 168-hour lookback and 24-hour horizon across GBM, LSTM, TCN, N-BEATS, TFT, and PatchTST. What it does not establish is the ceiling

performance of deep models under dataset scale-up or hyperparameter-intensive tuning beyond the early-stopping budget. The six-model comparison is therefore a fair head-to-head under realistic training constraints for this data regime, not a claim about the theoretical ceiling of any architecture. The finding is: *given this data and this training budget, feature engineering dominates architecture choice.* That is a useful and reproducible result for BESS dispatch practitioners.

11 Cross-Region and US Regime Analysis

11.1 Why US Direct Gains Are Weak

The weak US direct gains are a real result, not a formatting artifact. The locked US profile shows only 0.11% cost savings and 0.00% peak shaving even though the US stochastic run reports VSS = 297,092.71. Understanding why these facts coexist is analytically important.

11.2 Arbitrage Headroom and Tariff Structure

The first explanation is economic headroom. In the locked DE evaluation, the cost structure creates larger arbitrage and peak-shaving opportunities for the battery. In the locked US evaluation window, the direct B1-versus-B2 decision leverage appears much smaller. A region can still have large stochastic value because uncertainty-aware planning matters under its scenario construction, while showing weak realized direct savings against a grid-only baseline over the specific locked horizon.

11.3 Battery Utilization and Constraint Binding

The second explanation is control leverage. If battery power, reserve, or terminal-SOC constraints bind more often in US runs, the optimizer has less room to convert better forecasts into visibly better direct cost or peak outcomes. Positive VSS then means uncertainty matters inside the optimization, but the battery has too little practical freedom to turn that uncertainty value into large realized percentage improvements.

11.4 Calibration Quality and Optimization Leverage

The third explanation is calibration asymmetry. Under-coverage in some US targets, especially wind, implies that the optimizer sees less trustworthy intervals than the DE pipeline does. In that setting, the shield can still improve safety and maintain stochastic value, but direct economic improvements will be muted because the system spends more of its effort compensating for uncertainty than exploiting favorable operating windows.

11.5 Region Heterogeneity Inside "US"

The canonical evidence lock uses the dashboard US profile, which aggregates across three structurally different balancing authorities. The release-family assets describe MISO, PJM, and ERCOT separately, and the per-BA decomposition in Table 16 reveals that the aggregate US headline numbers mask substantial authority-specific variation.

11.6 Per-Balancing-Authority Dispatch Impact

Three findings emerge from the per-BA decomposition:

1. **PJM dominates aggregate value.** PJM accounts for the majority of the aggregate US cost savings (14.26% cost reduction) due to its wider price spreads between peak and off-peak hours and its heavier reliance on fossil generation. However, PJM's optimization trades cost

Table 16: Per-balancing-authority dispatch impact from locked release-family artifacts. All values from the optimized-forecast controller versus grid-only baseline over 168 test steps. PJM's large cost saving and negative carbon delta reflect its heavier fossil mix and wider price spreads.

BA	**Cost Δ (%)**	**Carbon Δ (%)**	**Peak Δ (%)**	**Test Steps**
MISO	+0.11	+0.11	0.00	168
ERCOT	+0.19	+0.28	0.00	168
PJM	+14.26	−0.57	0.00	168

for carbon: carbon increases by 0.57%, indicating that the optimizer shifts some load away from cleaner hours to exploit larger price differentials.

2. **MISO and ERCOT show modest symmetric gains.** Both achieve small positive cost and carbon reductions (0.11–0.28%) with no peak shaving, consistent with tighter operating margins and less exploitable price structure in the locked evaluation window.
3. **Peak shaving is zero across all BAs.** The battery capacity relative to system peak is too small to materially reduce peak demand in any US authority. This is a scale effect, not an algorithmic failure: the locked 100 MW/100 MWh battery against 50–100 GW system peaks has negligible peak-shaving leverage.

11.7 Per-Balancing-Authority Structural Differences

Although the canonical lock uses the MISO-derived US profile, the release-family assets reveal important structural differences across the three balancing authorities:

Table 17: Structural differences across US balancing authorities in the release-family assets.

Characteristic	MISO	PJM	ERCOT
Geography	Midwest (15 states)	Mid-Atlantic / Great Lakes	Texas (isolated grid)
Primary renewables	Wind-dominant	Mixed wind/solar	Wind + solar growing
Market structure	Day-ahead + real-time	Day-ahead + real-time	Day-ahead + real-time
Interconnection	Eastern	Eastern	Independent (ERCOT)
Load profile	Industrial + agricultural	Urban + industrial	Extreme summer peaks
Wind variability	High (Great Plains)	Moderate (coastal/inland)	High (West Texas)

These structural differences have direct implications for dispatch optimization and explain the per-BA variation in Table 16. PJM's heavier fossil mix creates wider price differentials for the optimizer to exploit, while ERCOT's isolated grid means import/export constraints bind differently. MISO's wind-dominant renewable profile makes renewable curtailment the primary battery value driver rather than price arbitrage.

The coexistence of weak aggregate US improvements and large US VSS is not contradictory; it reflects the difference between realized percentage gains and the internal value of uncertainty-aware planning. Over the locked US evaluation window, the B1-versus-B2 comparison offers limited arbitrage and peak-shaving headroom in MISO and ERCOT, so even better dispatch decisions translate into only small realized percentage changes. PJM's stronger results show that the architecture can realize large gains when market structure provides sufficient headroom. The

stochastic program still benefits from modeling uncertainty in all three BAs because that uncertainty affects reserve use, constraint satisfaction, and tail-risk exposure—even when realized direct savings remain small.

12 Governance and Reproducibility

12.1 Source of Truth

This paper uses a markdown-first and artifact-locked workflow. The canonical manuscript source is `paper/PAPER_DRAFT.md`. The corresponding LaTeX manuscript is this file, and the shorter conference cut is `paper/paper_r1.tex`. Quantitative claims are valid only if they appear in the manuscript and remain traceable to `paper/metrics_manifest.json`, `paper/claim_matrix.csv`, and the locked report artifacts.

12.2 Repo-to-Paper Traceability

Table 18: Repository artifact traceability for the paper pipeline.

Repo artifact family	Primary role in the paper	Main paper touchpoint
README + setup docs	Environment bootstrap, run commands, and operator setup	Governance and appendices
Train/eval code	Implements forecasting, uncertainty calibration, dispatch, and CPSBench evaluation	Methods and experimental protocol
Configs + seeds	Fix dataset scope, model settings, and randomization policy	Experimental protocol and appendix hyperparameters
Data + preprocessing	Define dataset construction, split logic, and feature generation	Background and experimental protocol
Logs + checkpoints	Preserve run IDs, solver outputs, metrics, and locked evidence values	Results and governance
Plotting + publication scripts	Build paper-facing figures, tables, and release assets	Results and appendices

12.3 Validation Gates

The release gates for this paper are intentionally concrete:

1. `python3 scripts/validate_paper_claims.py` must pass.
2. `python3 scripts/sync_paper_assets.py -check` must pass.
3. Core numbers, run IDs, and titles must stay aligned across markdown and LaTeX.

These gates are non-negotiable: if a draft passes review but fails validation, the release is blocked until the drift is resolved. The rationale is that in a long-running applied project with multiple artifact families, manual consistency checking is both tedious and error-prone. Automating the check ensures that the scientific claims in the paper cannot silently diverge from the evidence artifacts they cite.

12.4 What Governance Does Not Claim

The governance contribution is scoped to this paper. It does not claim a universal reproducibility framework for ML research. It does not claim that artifact locking solves all reproducibility challenges—only that, for a multi-artifact cyber-physical systems project with multiple balancing authorities and release families, automated claim locking is necessary to prevent manuscript-artifact drift. The community-level argument for stronger reproducibility standards is made by Pineau et al. [20]; this paper is a concrete instantiation of that argument in a specific CPS domain.

12.5 Incident Response and Operational Safeguards

Beyond static validation gates, the governance stack includes an operational response sequence for handling common runtime failures during evaluation:

1. **Stale data detection**: if upstream delay exceeds `expected_cadence_s` (3,600 s), the system blocks dispatch and holds current state.
2. **Forecast drift**: KS-test or Page-Hinkley triggers raise an alert; if drift persists beyond `cooldown_steps` (24 steps), the retraining workflow is invoked.
3. **Solver infeasibility**: if the optimizer returns infeasible status, DC^{3}S falls back to the grid-only baseline (B2) and logs the event.
4. **Certificate chain breaks**: if a certificate cannot be emitted (e.g., DuckDB write failure), the system enters conservative mode and disables further dispatch until the audit path is restored.

13 Discussion and Threats to Validity

13.1 Implications

The results support a systems-level claim: telemetry reliability should be treated as part of the control state, not as a peripheral monitoring variable. In this paper, the safety win comes not from maximum conservatism but from conditional conservatism. The controller widens uncertainty sets when telemetry quality deteriorates and stays less intrusive otherwise.

13.2 CPS Security Implications

The DC^{3}S architecture has broader implications for CPS security beyond the battery dispatch setting studied here. Traditional CPS security focuses on intrusion detection and network isolation [17]. DC^{3}S demonstrates an alternative, control-theoretic defense: rather than attempting to detect and block all adversarial inputs, the controller *adapts its uncertainty model* in response to degraded input quality. This defense-in-depth strategy has several properties that are relevant to CPS security more broadly:

1. **Graceful degradation**: Rather than switching between "secure" and "insecure" operating modes, the controller continuously adjusts its conservatism based on telemetry quality. This avoids the cliff-edge failure modes that binary security classifiers can produce.
2. **Attack-agnostic defense**: The reliability scoring does not attempt to classify the *cause* of degradation (equipment failure vs. adversarial manipulation). Instead, it measures the *statistical properties* of the telemetry (missingness, staleness, spikes, ordering) and adjusts accordingly. This makes the defense robust to novel attack vectors that might bypass signature-based detectors.

3. **Auditable response**: The certificate chain provides a forensic record of how the controller responded to each degradation event. In a security incident, this record enables post-hoc analysis of whether the shield's response was proportionate and whether any unsafe commands were attempted.

The limitation of this approach is that it cannot prevent data corruption that is *statistically indistinguishable* from clean data. An adversary who can inject small, consistent biases into the telemetry stream may evade the reliability scoring. This is a fundamental limitation of any statistical anomaly detector, not specific to DC^{3}S.

13.3 Deployment Considerations

Translating the software-in-the-loop results of this paper to production deployment requires addressing several practical concerns that the current evaluation does not cover:

Latency requirements. The DC^{3}S step (reliability scoring → inflation → solve → repair → certify) must complete within the control cycle time. At hourly dispatch resolution, the current implementation's <5% overhead is negligible. At sub-minute resolution, the solver call becomes the bottleneck, and warm-starting or constraint-screening optimizations would be needed.

BMS integration. The current `BatteryPlant` simulator uses idealized charge/discharge efficiency. Production BMS systems have state-of-health-dependent efficiency curves, cell-level imbalances, and temperature constraints that affect the feasible action set. Integrating DC^{3}S with a real BMS requires mapping the abstract SOC constraints to physical cell-level limits.

Certificate storage. At hourly resolution, the certificate chain grows by ∼8,760 records per year per device. At sub-minute resolution, the storage and query requirements become significant. The current DuckDB storage is suitable for single-device evaluation but would need to be replaced with a production time-series database for fleet-scale deployment.

Multi-device coordination. This paper evaluates a single battery. In a fleet setting, multiple devices share grid constraints (e.g., transformer capacity, feeder limits). Extending DC^{3}S to multi-device coordination requires either a centralized shield that projects the joint action vector or a distributed protocol where devices negotiate safety margins.

13.4 Negative and Neutral Findings

The negative and asymmetric findings are analytically important because they clarify where the method is genuinely strong, where the current evidence is limited, and how the observed behavior should be interpreted.

1. US peak shaving remains 0.00%.
2. Some target-regime combinations remain under-covered at the base calibration layer.
3. GBM dominates the locked deep baselines.
4. Cross-region impact magnitudes vary sharply despite the shared architecture.

13.5 Threats to Internal Validity

The main internal-validity risk is manuscript-artifact drift. This is mitigated by run IDs, manifest locks, and validator checks, but not eliminated permanently. Another internal-validity risk is that some interpretive claims in the diagnosis sections are based on locked artifacts and configuration evidence rather than fresh reruns targeted specifically at those hypotheses.

13.6 Threats to External Validity

This paper is intentionally scoped to the locked DE and US windows. Generalization to different tariff regimes, different battery sizes, or different telemetry infrastructures remains future work. The multi-BA US evidence surface improves external validity, but it does not convert this paper into a universal market study.

13.7 Threats to Construct and Conclusion Validity

Construct validity depends on the proxy cost and carbon signals being appropriate enough for comparative analysis. Conclusion validity depends on scenario design and calibration definitions remaining stable. The formal results (Theorems 1 to 3) should not be overstated: Theorem 1 gives marginal coverage preservation under inflation; Theorem 2 gives group-conditional coverage under a Mondrian partition; and Theorem 3 gives one-step SOC invariance under a tightened tube. Together they formalize why the DC^3S construction is correct, but they do not establish closed-loop stability, optimal inflation, or finite-time convergence.

The theorems should be read as correctness conditions for the safety shield's construction, not as a standalone contribution in conformal prediction theory. They do not prove that the chosen inflation law is optimal among all monotone laws, do not guarantee minimum intervention rate, and do not remove the need for empirical validation. The practical safety of the system depends on how reliability scoring, drift behavior, forecast quality, optimization, and action repair interact under realistic faults. The empirical evidence in Section 9.2 and the ablation study in Section 9.6 provide the primary safety validation. Accordingly, the paper's contribution is systems-oriented: the theorems validate the construction, while the locked benchmark results validate the method.

14 Limitations and Future Work

14.1 Current Limitations

1. Evidence is limited to locked DE and US windows.
2. Evaluation is software-in-the-loop rather than hardware field validation; CHIL, HIL, and field pilots remain explicit future work and are not claimed in the present evidence base. All safety and economic claims in this manuscript rest on artifact-locked simulation evidence; physical battery control, inverter integration, and BMS-level constraint enforcement are outside the current evidence lock.
3. Cost and carbon signals are engineering proxies rather than full settlement models.
4. The formal results (Theorems 1 to 3) formalize correctness of the RAC-Cert construction and safety-margin monotonicity but do not establish closed-loop stability or optimal inflation.
5. The six-model forecasting comparison is bounded to the present data regime and artifact state. Larger datasets, weaker tabular preprocessing, or longer-context forecasting could shift the architecture ranking.

14.2 Near-Term Future Work

Near-term work should focus on three items: decomposing the US narrative more explicitly into MISO/PJM/ERCOT in the main text, improving under-covered target-regime calibration before dispatch, and adding a deployment-facing latency/runtime benchmark together with a publication-grade case trace showing telemetry degradation, interval widening, action repair, and safe dispatch behavior over time.

14.3 Longer-Term Future Work

Longer-term work includes hardware-in-the-loop validation with physical inverter and BMS integration, richer market constraints that capture settlement rules more faithfully, conditional-coverage analysis under non-stationary degradation, and stronger release automation with cryptographically signed evidence bundles. A further research direction is extending the RAC-Cert framework to multi-agent dispatch settings where multiple batteries share a common telemetry infrastructure and must coordinate safety constraints. Those hardware-facing steps are intentionally deferred beyond the current evidence lock.

15 Conclusion

Battery dispatch controllers can appear safe when evaluated on observed state while simultaneously violating physical battery limits on true state — a hidden failure mode that standard evaluation protocols do not expose. This paper quantifies that gap at 3.9% true-SOC violation rate (P95 severity 333 MWh) for deterministic dispatch under realistic telemetry faults, and shows that it is invisible to observed-state evaluation. We introduced CPSBench to make this gap measurable, and DC^{3}S to close it.

DC^{3}S achieves zero true-SOC violations at a 2.8% intervention rate — the shield is surgically conservative, not blanket conservative. It activates only when telemetry reliability deteriorates, leaving 97.2% of dispatch steps unmodified. Cost-side results confirm that safety and economic performance are not fundamentally at odds: DC^{3}S delivers 7.11% cost savings, 0.30% carbon reduction, and 6.13% peak shaving on the locked DE evaluation, with positive stochastic value in both DE (VSS = 2,708.61) and US (VSS = 297,092.71) regions. US direct gains are modest (0.11% cost, 0.00% peak) — a regime-dependent result that reflects tighter operating margins and compressed arbitrage headroom across the three US balancing authorities, not a failure of the approach.

The promoted six-model forecasting comparison points in one clear direction: GBM remains the strongest operational model under a fair equal-budget protocol on these moderate-sized, feature-rich grid datasets. The practical implication for BESS practitioners is clear: invest in feature engineering before model architecture on data of this type.

The core message. Trustworthy grid-scale dispatch requires treating telemetry reliability as a first-class input to uncertainty calibration. The measurement channel is not an implementation detail — it is a safety-critical component. When that channel degrades, the controller's uncertainty model must degrade with it, not remain frozen at calibration-time assumptions. DC^{3}S operationalizes this principle in a single online loop with per-step auditable evidence. The hidden safety gap is real, measurable in software-in-the-loop evaluation, and closeable — validating these findings on physical battery hardware is the natural next step.

The conclusion-level locked numeric summary is therefore explicit: 7.11% DE cost savings, 0.30% DE carbon reduction, 6.13% DE peak shaving, 0.11% US cost savings, 0.13% US carbon reduction, and 0.00% US peak shaving.

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A Replication Checklist

All main claims should be reproducible from the following steps:

```
# Validate paper claims against locked artifacts
python3 scripts/validate_paper_claims.py
python3 scripts/sync_paper_assets.py --check

# Train all models and build uncertainty artifacts
make train-all-ba

# Run CPSBench fault sweep
make r1-cpsbench

# Run ablation study and publication checks
make r1-full
make r1-verify
make paper-sync
```

Canonical run IDs: DE `20260217_165756`, US `20260217_182305`. Dashboard profile lock timestamp: `2026-02-17T11:15:38.623283`. All artifacts are hash-tracked in the release manifest.

B Hyperparameters and Configuration

Table 19: Core training and RAC-Cert hyperparameters.

Group	Parameter	Value	Source
train_forecast_de	horizon_hours	24	configs/train_forecast.yaml
train_forecast_de	lookback_hours	168	configs/train_forecast.yaml
train_forecast_de	cross_validation.n_folds	10	configs/train_forecast.yaml
train_forecast_de	cross_validation.gap	24	configs/train_forecast.yaml
train_forecast_us	horizon_hours	24	configs/train_forecast_eia930.yaml
train_forecast_us	lookback_hours	168	configs/train_forecast_eia930.yaml
train_forecast_us	cross_validation.n_folds	10	configs/train_forecast_eia930.yaml
train_forecast_us	cross_validation.gap	24	configs/train_forecast_eia930.yaml
dc3s_rac	sensitivity_probe	heuristic	configs/dc3s.yaml
dc3s_rac	sens_eps_mw	25.0	configs/dc3s.yaml
dc3s_rac	sens_norm_ref	0.5	configs/dc3s.yaml
dc3s_rac	beta_reliability	0.7	configs/dc3s.yaml
dc3s_rac	beta_sensitivity	0.5	configs/dc3s.yaml
dc3s_rac	k_sensitivity	0.4	configs/dc3s.yaml
regime_cqr	enabled	True	configs/uncertainty.yaml
regime_cqr	n_bins	3	configs/uncertainty.yaml
regime_cqr	vol_window	24	configs/uncertainty.yaml
regime_cqr	quantile_backend_policy	strict	configs/uncertainty.yaml

Key configuration values from the locked `configs/` directory:

- Forecast: 24-hour horizon, 168-hour lookback, 10-fold time-aware CV with 24-hour gap.
- Conformal: $\alpha = 0.10$, volatility binning window = 168 hours.

- RAC-Cert: $\kappa_r = 0.2$ (quality), $\kappa_d = 0.0$ (drift, disabled), $\kappa_s = 0.4$ (sensitivity), `infl_max` = 2.0, shrinkage $\tau = 30.0$.
- Reliability: $\lambda_{\text{delay}} = 0.002$, $\beta_{\text{spike}} = 0.25$, $\gamma_{\text{ooo}} = 0.35$, $w_{\min} = 0.05$.
- Drift detection: Page-Hinkley with $\delta = 0.01$, $\lambda = 5.0$, warmup 48 steps, cooldown 24 steps.
- Shield: projection mode, reserve SOC under drift = 8%.

Table 20: Optimization configuration from `configs/optimization.yaml`.

Parameter	Value	Role
Objective weights		
Cost weight (λ_c)	1.0	Energy cost coefficient
Carbon weight (λ_g)	1.2	Carbon-weighted cost coefficient
Battery parameters		
Capacity	20,000 MWh	Rated energy capacity ($E_{\max}$)
Max power	5,000 MW	Charge/discharge power limit ($P_{\max}$)
Efficiency	0.92	Round-trip charge/discharge efficiency
Degradation cost	\$10/MWh	Cycle-aging penalty per throughput
Constraint penalties		
Curtailment	\$500/MW	Penalty for wasted renewable energy
Unmet load	\$10,000/MW	High penalty (should not trigger)
Peak	\$1/MW	Small peak-demand penalty
Robust dispatch		
Risk weight (worst-case)	0.35	DRO blend: 1.0 = pure worst-case
Min SOC	1,000 MWh	Minimum allowed SOC
Initial SOC	10,000 MWh	Starting battery state (50%)

C Data Assets and Feature Design

Table 21: Feature family composition by region.

Feature family	DE (count)	US (count)
Weather	40	56
Lag	36	42
Calendar	6	6
Total	**94**	**114**

Table 22: Target distribution snapshot (DE).

Target	Mean	Std	Min	Max	Median
Load (MW)	55,156	9,998	31,923	76,925	54,752
Wind (MW)	14,369	10,322	136	46,064	11,728
Solar (MW)	5,013	7,666	0	32,947	156

The feature design follows a leakage-safe temporal framing: all lag and rolling features use strictly past data relative to the forecast origin, with a 24-hour gap between training and test folds. Weather features include temperature, humidity, precipitation, cloud cover, and wind

Table 23: Target distribution snapshot (US/MISO).

Target	Mean	Std	Min	Max	Median
Load (MW)	75,757	11,819	52,940	120,343	73,564
Wind (MW)	10,956	6,042	0	26,132	10,184
Solar (MW)	2,875	3,843	0	14,315	110

speed at multiple heights. Calendar features encode hour-of-day, day-of-week, month, weekend indicator, and season. The lag structure covers 1-hour, 24-hour, and 168-hour (weekly) lags for each target, plus rolling mean and standard deviation at 24-hour and 168-hour windows.

The US feature set is larger (114 vs. 94) primarily because the US pipeline includes additional lagged exogenous variables (price, carbon intensity at multiple lags) and more granular weather features from the expanded geographic coverage of three balancing authorities.

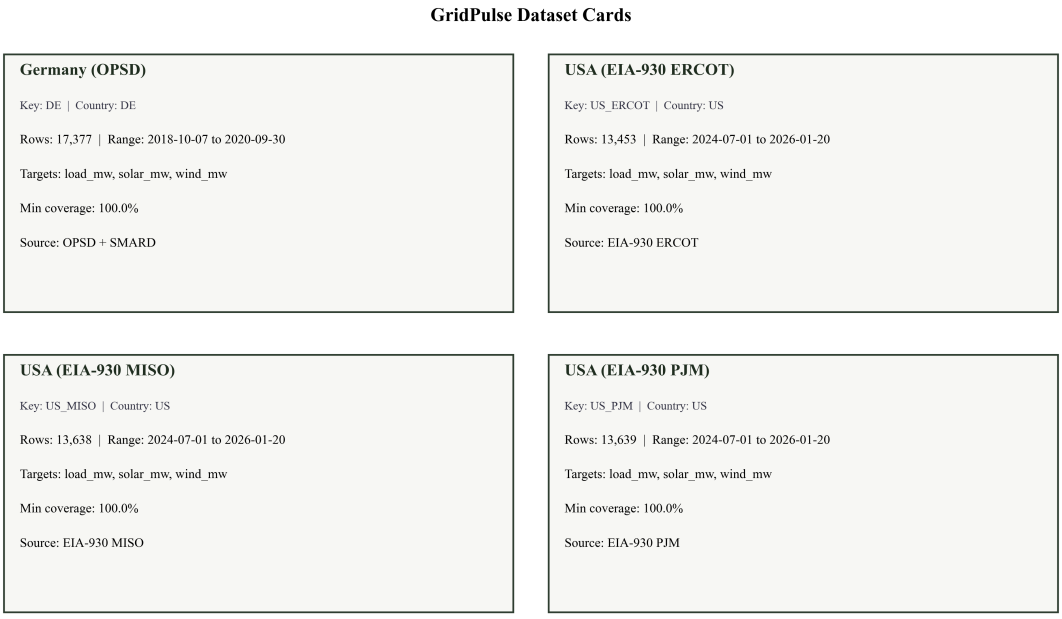

Figure 10: Dataset cards for all regions in the release scope.

D Additional Figures

E Artifact Inventory

F Operational Failure Modes

G Notation and Symbols

H Extended Algorithm Details

This appendix provides the full algorithmic specification for two key DC^{3}S subsystems that are summarized in the main text: the FTIT state update and the reliability scoring function.

These algorithms are presented in a form that matches the actual implementation in `src/gridpulse/dc3s/ft` and `src/gridpulse/dc3s/quality.py` respectively. The default parameter values correspond to the locked configuration in `configs/dc3s.yaml` (Table 5).

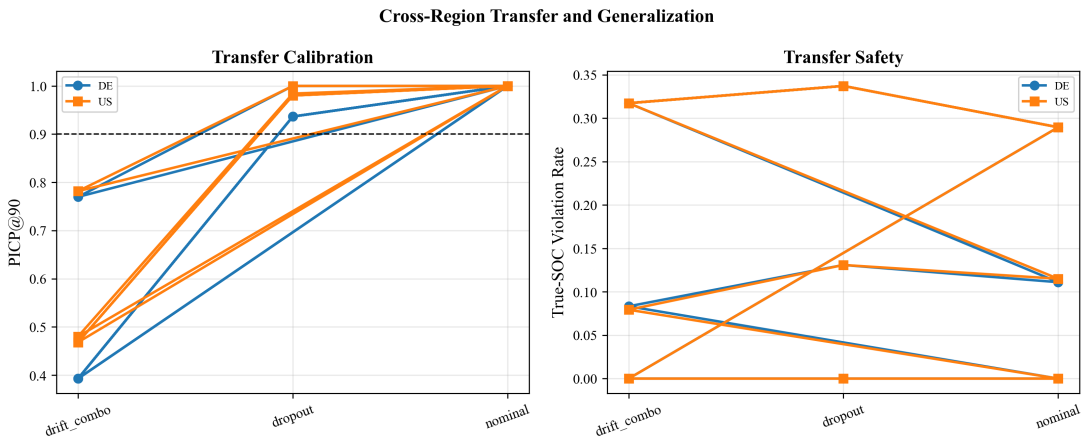

Figure 11: Cross-region transfer and generalization performance summary.

Table 24: Core manuscript and evidence artifacts.

Artifact	Purpose
`paper/PAPER_DRAFT.md`	Canonical thesis manuscript source
`paper/paper.tex`	Thesis LaTeX twin of the markdown source
`paper/paper_r1.tex`	Shorter conference derivative
`paper/metrics_manifest.json`	Canonical metric lock and display contract
`paper/claim_matrix.csv`	Claim-level provenance and status tracking
`scripts/validate_paper_claims.py`	Cross-file claim validation
`scripts/sync_paper_assets.py`	Paper-asset synchronization audit
`reports/impact_summary.csv`	Locked DE impact values
`reports/eia930/impact_summary.csv`	Locked US impact values
`reports/research_metrics_de.csv`	DE stochastic summary rows
`reports/research_metrics_us.csv`	US stochastic summary rows
`data/dashboard/manifest.json`	Profile lock metadata
`configs/dc3s.yaml`	DC^{3}S configuration (Table 5)
`configs/optimization.yaml`	Optimization parameters (Table 20)

Table 25: Operational failure modes and safety controls.

Failure Mode	Trigger	Detection	Mitigation	Fallback
Missing/stale data	Upstream delay	Readiness checks	Block unsafe dispatch	Hold current state
Forecast drift	Distribution shift	KS/model monitors	Retraining workflow	Conservative mode
Solver infeasible	Extreme intervals	Solver status flags	Safe infeasible response	Grid-only baseline
Unsafe command	SOC/power violation	BMS validation	Reject with detail	Maintain safe state
Control degradation	Heartbeat timeout	Watchdog detection	Lock remote control	Manual/local
Certificate failure	DuckDB write error	Audit path check	Conservative mode	Disable dispatch

Table 26: Principal notation used throughout this paper.

Symbol	Definition
State and telemetry	
$\mathbf{z}_t$	Clean (true) telemetry vector at time t
$\tilde{\mathbf{z}}_t$	Degraded (observed) telemetry after $\mathcal{D}$
$\mathcal{D}$	Degradation operator (dropout, delay, spikes, reordering)
$\mathrm{SOC}_t^{\mathrm{true}}$	True state of charge (MWh)
$\mathrm{SOC}_t^{\mathrm{obs}}$	Observed state of charge (MWh)
Battery	
$E_{\max}$	Rated battery energy capacity (MWh)
$P_{\max}$	Maximum charge/discharge power (MW)
η_c, η_d	Charge and discharge efficiency
$R_{\max}$	Maximum ramp rate (MW/step)
a_t	Dispatch action at step t (MW, positive = discharge)
$\mathcal{A}_t$	Feasible action set at step t
Forecasting and uncertainty	
$\hat{C}_t(\alpha)$	Base conformal prediction interval at level α
$\hat{C}_t^{\mathrm{RAC}}(\alpha)$	RAC-Cert inflated interval
α	Nominal miscoverage level (default 0.10)
$\hat{q}$	Empirical conformal quantile
DC^{3}S parameters	
w_t	Telemetry reliability score $\in [w_{\min}, 1]$
d_t	Drift detection flag (binary)
s_t	Normalized sensitivity signal $\in [0, 1]$
$\kappa_r, \kappa_d, \kappa_s$	Inflation coefficients for quality, drift, sensitivity
$\varepsilon_{\mathrm{sens}}$	Sensitivity probe perturbation (MW)
τ_{shrink}	Shrinkage time constant (steps)
Objective and economics	
$\lambda_c, \lambda_g, \lambda_p, \lambda_d$	Cost, carbon, peak, and degradation weights
c_t, g_t	Electricity price and carbon intensity at step t
L_t	Net load (load minus renewables) at step t
VSS	Value of the stochastic solution
EVPI	Expected value of perfect information
FTIT state	
γ_d	Exponential decay factor for fault rates (0.98)
$p_t^{(k)}$	Rolling empirical fault rate for fault k
γ_{MW}	SOC tube width (MW)
e_t	Accumulated SOC margin (MWh)
σ_t^2	Exponentially weighted residual variance

Algorithm 2 FTIT state update at each dispatch step

Require: Fault flags $\{f_t^{(k)}\}_{k=1}^{5}$, previous state $(n_t, s_t, \sigma_t^2, e_t)$, config $(\gamma_d, \alpha_k, \gamma_{\min}, \gamma_{\max}, \gamma_{\text{pow}})$
Ensure: Updated state $(n_{t+1}, s_{t+1}, w_t^{\text{FTIT}}, \sigma_{t+1}^2, e_{t+1}, \text{soc_tube})$

$n_{t+1} \leftarrow \gamma_d \cdot n_t + 1$
for each fault type k **do**
  $s_{t+1}^{(k)} \leftarrow \gamma_d \cdot s_t^{(k)} + \mathbb{1}[f_t^{(k)}]$
  $p_{t+1}^{(k)} \leftarrow s_{t+1}^{(k)} / \max(n_{t+1}, \epsilon)$
end for
$w_t^{\text{FTIT}} \leftarrow \prod_k [\text{clip}(1 - p_{t+1}^{(k)}, \epsilon, 1)]^{\alpha_k}$
$\sigma_{t+1}^2 \leftarrow \max(\sigma_{\text{floor}}^2,\ \gamma_d \cdot \sigma_t^2 + (1 - \gamma_d) \cdot \sigma_{\text{obs}}^2)$ ▷ residual variance
$\gamma_{\text{MW}} \leftarrow \gamma_{\min} + (1 - w_t^{\text{FTIT}})^{\gamma_{\text{pow}}} \cdot (\gamma_{\max} - \gamma_{\min})$
$e_{t+1} \leftarrow \text{clip}(\gamma_d \cdot e_t + \Delta t \cdot \gamma_{\text{MW}},\ e_{\min},\ e_{\max})$
$\text{soc_tube} \leftarrow [\text{SOC}_{\min} + e_{t+1},\ \text{SOC}_{\max} - e_{t+1}]$
return $(n_{t+1}, s_{t+1}, w_t^{\text{FTIT}}, \sigma_{t+1}^2, e_{t+1}, \text{soc_tube})$

Algorithm 3 Reliability scoring from raw telemetry

Require: Telemetry event $\tilde{\mathbf{z}}_t$, previous event $\tilde{\mathbf{z}}_{t-1}$, cadence Δ_{exp}
Ensure: Reliability score w_t, component penalties

$p_{\text{miss}} \leftarrow \max(0, 1 - \text{frac_missing}(\tilde{\mathbf{z}}_t))$
$\delta_s \leftarrow |t_{\text{arrive}} - t_{\text{expected}}|$ ▷ staleness in seconds
$p_{\text{delay}} \leftarrow \exp(-\lambda_{\text{delay}} \cdot \max(0, \delta_s))$ ▷ $\lambda = 0.002$
$p_{\text{ooo}} \leftarrow \begin{cases} 1 - \gamma_{\text{ooo}} & \text{if out-of-order detected} \\ 1 & \text{otherwise} \end{cases}$ ▷ $\gamma = 0.35$
$r_{\text{spike}} \leftarrow \max_j |\tilde{z}_{t,j} - \tilde{z}_{t-1,j}| / (|\tilde{z}_{t-1,j}| + \epsilon)$ ▷ relative jump
$p_{\text{spike}} \leftarrow \begin{cases} 1 & \text{if } r_{\text{spike}} \le \beta_{\text{spike}} \\ 1/(1 + r_{\text{spike}} - \beta_{\text{spike}}) & \text{otherwise} \end{cases}$ ▷ $\beta = 0.25$
$w_{\text{raw}} \leftarrow p_{\text{miss}} \cdot p_{\text{delay}} \cdot p_{\text{ooo}} \cdot p_{\text{spike}}$
$w_t \leftarrow \max(w_{\min}, \min(1, w_{\text{raw}}))$ ▷ $w_{\min} = 0.05$
return w_t, $(p_{\text{miss}}, p_{\text{delay}}, p_{\text{ooo}}, p_{\text{spike}})$